Linear Algebra

Author: Star
Email: wusida@westlake.edu.cn

Instructors: Prof. Yabin Zhang
Dr. Sean Brendon Lynch
Emails: zhangyabin@westlake.edu.cn
sblynch@westlake.edu.cn

Contents

Chapter 1 Vector Spaces

1 Definitions

Definition 1.1.1 A **field** $\mathcal{K}$ is a set equipped with addition and multiplication, denoted by $(\mathcal{K}, +, \cdot)$, satisfying the following axioms:

1. **Associative Laws**: let $a, b, c \in \mathcal{K}$,
$$a + b + c = a + (b + c), \qquad a \cdot b \cdot c = a \cdot (b \cdot c).$$
2. **Existence of Identity Elements**: let $a \in \mathcal{K}$,
$$a + 0_{\mathcal{K}} = a, \qquad a \cdot 1_{\mathcal{K}} = a,$$
where $0_{\mathcal{K}}$ and $1_{\mathcal{K}}$ is called the additive identity and multiplicative identity.
3. **Existence of Negative**: for any $a \in \mathcal{K}$, there exists one element in $\mathcal{K}$ denoted by $(-a)$ such that
$$a + (-a) = 0_{\mathcal{K}}.$$
4. **Existence of Reciprocal**: for any $a \in \mathcal{K}$ and $a \neq 0_{\mathcal{K}}$, there exists one element in $\mathcal{K}$ denoted by a^{-1} such that
$$a \cdot a^{-1} = 1_{\mathcal{K}}.$$
5. **Commutative Laws**: let $a, b \in \mathcal{K}$,
$$a + b = b + a, \qquad a \cdot b = b \cdot a.$$
6. **Distributive Law**: let $a, b, c \in \mathcal{K}$,
$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark 1.1.2 Closure is contained in the defined addition and multiplication.

Notation 1.1.3 For $a, b \in \mathcal{K}$, we usually write $a - b$ to replace $a + (-b)$ and write $\frac{a}{b}$ or a/b to replace $a \cdot b^{-1}$ when $b \neq 0_{\mathcal{K}}$.

Theorem 1.1.4 (Cancellation Law) Let $a, b, c \in \mathcal{K}$, if $a + c = b + c$, then $a = b$.

Corollary 1.1.5 $0_\mathcal{K}$ and $1_\mathcal{K}$ in **Field Axiom 2** is unique.

Example 1.1.6 $\mathbb{C}, \mathbb{R}, \mathbb{Q}$ are fields, but $\mathbb{N}$ is not a field because $2 \in \mathbb{N}$, but its reciprocal $\dfrac{1}{2} \notin \mathbb{N}$. This is the main difference between $\mathbb{Q}$ and $\mathbb{N}$.

Definition 1.1.7 Let $\mathcal{F}$ be a field. A subset $\mathcal{K} \subseteq \mathcal{F}$ is called a **subfield** of $\mathcal{F}$ if $\mathcal{K}$ is itself a field under the same operations of addition and multiplication as $\mathcal{F}$. Equivalently, $\mathcal{K}$ is a subfield of $\mathcal{F}$ if and only if the following conditions hold:

1. **Closure under Addition and Multiplication:** If x, y are elements of $\mathcal{K}$, then $x + y \in \mathcal{K}$ and $x \cdot y \in \mathcal{K}$.
2. **Existence of Additive and Multiplicative Inverses:** If $x \in \mathcal{K}$, then $(-x) \in \mathcal{K}$. If furthermore $x \neq 0$, $x^{-1} \in \mathcal{K}$.
3. **Existence of Identity Elements:** $0_\mathcal{K}$ and $1_\mathcal{K}$ are in $\mathcal{K}$.

Remark 1.1.8 Definition 1.1.7 provides a simpler way to prove a set to be a field. The third rule ensures that $\varnothing$ is not a field.

Proposition 1.1.9 Let $\mathcal{K}$ be a field and A, B are subfields of $\mathcal{K}$, then $A \cap B$ is also a subfield of $\mathcal{K}$.

Proposition 1.1.10 Let $\mathcal{K}$ be a field and A, B are subfields of $\mathcal{K}$, then $A \cup B$ is also a subfield of $\mathcal{K}$ if and only if $A \subseteq B$ or $B \subseteq A$.

Remark 1.1.11 For any elements in a field, we call them **number**s or **scalar**s.

Definition 1.1.12 A **vector space** V over a field $\mathcal{K}$ is a set equipped with two operations: vector addition and scalar multiplication, denoted by $(V, +, \cdot)$, satisfying the following axioms:

1. **Associative Laws:** for any $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in V$,
$$\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}.$$
2. **Existence of Identity Elements:** there exists a zero vector $\mathbf{0} \in V$ such that for any $\boldsymbol{v} \in V$,
$$\boldsymbol{v} + \mathbf{0} = \boldsymbol{v}.$$
3. **Existence of Additive Inverses:** for any $\boldsymbol{v} \in V$, there exists an element $-\boldsymbol{v} \in V$ such that
$$\boldsymbol{v} + (-\boldsymbol{v}) = \mathbf{0}.$$
4. **Commutative Law:** for any $\boldsymbol{u}, \boldsymbol{v} \in V$,
$$\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}.$$
5. **Distributive Laws:** for all $a, b \in \mathcal{K}$ and $\boldsymbol{u}, \boldsymbol{v} \in V$,
$$a \cdot (\boldsymbol{u} + \boldsymbol{v}) = a \cdot \boldsymbol{u} + a \cdot \boldsymbol{v}, \quad (a + b) \cdot \boldsymbol{v} = a \cdot \boldsymbol{v} + b \cdot \boldsymbol{v}.$$

6. **Compatibility of Scalar Multiplication:** for all $a, b \in \mathcal{K}$ and $\boldsymbol{v} \in V$,

$$(ab) \cdot \boldsymbol{v} = a \cdot (b \cdot \boldsymbol{v}).$$

7. **Identity of Scalar Multiplication:** for all $\boldsymbol{v} \in V$,

$$1_{\mathcal{K}} \cdot \boldsymbol{v} = \boldsymbol{v}.$$

Theorem 1.1.13 The zero vector $\mathbf{0}$ is unique.

Definition 1.1.14 Let V be a vector space over a field $\mathcal{K}$. A non-empty subset $W \subseteq V$ is called a **subspace** of V if W is itself a vector space over $\mathcal{K}$ under the same operations of addition and scalar multiplication as V. Equivalently, W is a subspace of V if and only if the following conditions hold for all $\boldsymbol{u}, \boldsymbol{v} \in W$ and $a \in \mathcal{K}$:

1. **Closure under Addition:**

$$\boldsymbol{u} + \boldsymbol{v} \in W.$$

2. **Closure under Scalar Multiplication:**

$$a \cdot \boldsymbol{u} \in W.$$

Notation 1.1.15 If W is a subspace of V, we write $W \leq V$.

Remark 1.1.16 We state W to be a non-empty set at first so we needn't to say $\mathbf{0} \in W$.

Remark 1.1.17 For any elements in a vector space, we call them **vector**s.

Example 1.1.18 If $\mathcal{F}$ is a field, then $\mathcal{F}^n$ is always a vector space over $\mathcal{F}$. Furthermore, $\mathcal{F}^n$ is a vector space over $\mathcal{K}$ whenever $\mathcal{K} \subseteq \mathcal{F}$ is a subfield of $\mathcal{F}$.

Definition 1.1.19 Let V be a vector space over a field $\mathcal{K}$, and let vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n \in V$. Any vector of the form

$$\boldsymbol{v} = a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n,$$

where $a_1, a_2, \ldots, a_n \in \mathcal{K}$, is called a **linear combination** of $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$.

Definition 1.1.20 Let V be a vector space over a field $\mathcal{K}$, and let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\} \subseteq V$. The set of all linear combinations of vectors in S is called the **span** of S, denoted by

$$\operatorname{span}(S) = \{a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n \mid a_1, a_2, \ldots, a_n \in \mathcal{K}\}.$$

Theorem 1.1.21 The span of S is the smallest subspace of V containing S. Particularly, $\operatorname{span}(\varnothing) = \{\mathbf{0}\} = \operatorname{span}\{\mathbf{0}\}$.

Definition 1.1.22 Let V be a vector space over a field $\mathcal{K}$. A subset $S \subseteq V$ is called a **spanning set** of V or a set of **generators** if $\operatorname{span}(S) = V$, that is, every vector in V can be expressed as a linear combination of finitely many vectors from S.

Definition 1.1.23 Let V be a vector space over a field $\mathcal{K}$, and let vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n \in V$. The set $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ is said to be **linearly dependent** if there exist scalars $a_1, a_2, \ldots, a_n \in \mathcal{K}$, not all zero, such that

$$a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n = \mathbf{0}.$$

Otherwise, if this equation holds only when $a_1 = a_2 = \cdots = a_n = 0$, the set is said to be **linearly independent**.

Remark 1.1.24 $\{\mathbf{0}\}$ is linearly dependent, but $\varnothing$ is linearly independent.

2 Bases

Definition 1.2.1 Let V be a vector space over a field $\mathcal{K}$. A subset $\mathcal{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\} \subseteq V$ is called a **basis** of V if it satisfies the following two conditions:

1. $\mathcal{B}$ is linearly independent.
2. $\mathcal{B}$ spans V, i.e. $\operatorname{span}(\mathcal{B}) = V$.

Theorem 1.2.2 Every vector in V can be written uniquely as a linear combination of the vectors in $\mathcal{B}$.

Remark 1.2.3 Though we use set notation to denote a basis, the order *does* matter. For instance, two bases

$$\mathcal{B}_1 = \{(1,0),(0,1)\}, \quad \mathcal{B}_2 = \{(0,1),(1,0)\}$$

are considered to be different. You will know why when we discuss about matrix representation of linear maps.

Theorem 1.2.4 Let $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ be a set of elements of a vector space V over a field $\mathcal{K}$, and let r be a positive integer with $r \leq n$. We say that $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_r\}$ is a **maximal subset of linearly independent elements** if $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r$ are linearly independent, and moreover, for any $\boldsymbol{v}_i$ with $i > r$, the set $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r, \boldsymbol{v}_i\}$ is linearly dependent.
If, in addition, $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ is a set of generators of V, then the set $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_r\}$ is a **basis** of V.

3 Dimension

Theorem 1.3.1 Let V be a vector space over a field $\mathcal{K}$. Let $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_m\}$ be a basis of V over $\mathcal{K}$. Let $\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_n$ be elements of V, and assume that $n > m$. Then $\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_n$ are linearly dependent.

Theorem 1.3.2 Let V be a vector space over a field $\mathcal{K}$, and suppose that one basis of V has n elements and another basis has m elements. Then $m = n$.

Remark 1.3.3 Theorem 1.3.2 ensures that the dimension of a finite-dimensional vector space is well-defined, that is, it does not depend on the choice of basis.

Definition 1.3.4 Let V be a vector space over a field $\mathcal{K}$. If V has a finite basis $\mathcal{B}$ consisting of n vectors, then V is said to be **finite-dimensional**, and the number n is called the **dimension** of V, denoted by $\dim(V) = n$.
If no finite basis exists, V is said to be **infinite-dimensional**.

Remark 1.3.5 The existence of a basis in a finite-dimensional vector space can be proved via Steinitz Exchange Lemma. However, the existence of a basis in infinite-dimensional vector space relies on the Axiom of Choice.

Example 1.3.6 Let $\mathcal{K}$ be a field. Define

$$V = \mathcal{K}[x] = \left\{ a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \;\middle|\; n \in \mathbb{N},\ a_i \in \mathcal{K} \right\},$$

the set of all polynomials with coefficients in $\mathcal{K}$. Then V is a vector space over $\mathcal{K}$ with the usual polynomial addition and scalar multiplication.
The set

$$\{\, 1, x, x^2, x^3, \dots \}$$

is linearly independent, and no finite subset of it spans V. Hence $\mathcal{K}[x]$ is an **infinite-dimensional** vector space over $\mathcal{K}$.

Definition 1.3.7 Let V be a vector space over a field $\mathcal{K}$. A subspace of V with dimension 1 is called a **line** in V. A subspace of V with dimension 2 is called a **plane** in V.

Definition 1.3.8 Let V be a finite-dimensional vector space over a field $\mathcal{K}$, and let $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \dots, \boldsymbol{e}_n\}$ be an ordered basis of V. For any vector $\boldsymbol{v} \in V$, there exist unique scalars $a_1, a_2, \dots, a_n \in \mathcal{K}$ such that

$$\boldsymbol{v} = a_1 \boldsymbol{e}_1 + a_2 \boldsymbol{e}_2 + \cdots + a_n \boldsymbol{e}_n.$$

The scalars $a_1, a_2, \dots, a_n$ are called the **components** or **coordinates** of $\boldsymbol{v}$ with respect to the basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \dots, \boldsymbol{e}_n\}$. And the n-tuple $A = (a_1, \dots, a_n)$ is called the **coordinate vector** of **v**.

Definition 1.3.9 Let V be a vector space over a field $\mathcal{K}$, and let $S \subseteq V$. A subset $\mathcal{B} \subseteq S$ is called a **maximal linearly independent subset** of S if

1. $\mathcal{B}$ is linearly independent;
2. for any vector $\boldsymbol{v} \in S \setminus \mathcal{B}$, the set $\mathcal{B} \cup \{\boldsymbol{v}\}$ is linearly dependent.

Theorem 1.3.10 Let V be a vector space over a field $\mathcal{K}$, and let the set $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ be a maximal set of linearly independent elements of V. Then $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ is a basis of V.

Theorem 1.3.11 Let V be a vector space over a field $\mathcal{K}$ of dimension n, and let $\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n$ be linearly independent elements of V. Then $\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n$ constitute a basis of V.

Proposition 1.3.12 Let V be a vector space over a field $\mathcal{K}$, and let W be a subspace of V. If $\dim W = \dim V$, then $V = W$.

Remark 1.3.13 This proposition gives us a way to prove that two sets (vector space) are the same.

Proposition 1.3.14 Let V be a vector space over a field $\mathcal{K}$ of dimension n. Let r be a positive integer with $r < n$, and let $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_r$ be linearly independent elements of V. Then there exist elements $\boldsymbol{v}_{r+1}, \boldsymbol{v}_{r+2}, \ldots, \boldsymbol{v}_n$ in V such that

$$\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$$

is a basis of V.

Remark 1.3.15 These theorems and propositions all point to the same idea: a linearly independent set may be too small to span the whole vector space, while a spanning set may be too large to be linearly independent. A basis lies exactly at the boundary between these two situations. It is both a minimal spanning set and a maximal linearly independent set. In this sense, it plays a role somewhat analogous to a supremum.

Theorem 1.3.16 Let V be a vector space over a field $\mathcal{K}$ having a basis consisting of n elements. Let W be a subspace of V that does not consist of the zero vector alone. Then W has a basis, and the dimension of W satisfies

$$\dim W \leq n.$$

Chapter 2 Matrices

These definitions are just for a quick review. Please refer to Artin's Algebra Chapter 1.1-1.3 for more detailed information.

1 Definitions

Definition 2.1.1 Let $\mathcal{K}$ be a field and let m, n be two positive integers. An array of elements in $\mathcal{K}$ of the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

is called a **matrix** over $\mathcal{K}$ of size $m \times n$, or an m-by-n matrix.

Definition 2.1.2 For convenience, we may write $A = (a_{ij})$ for $1 \leq i \leq m$ and $1 \leq j \leq n$ to denote a matrix in $\mathcal{K}^{m\times n}$. The element a_{ij} is called the (i, j)-entry of A. Each **column** of A is a vector in $\mathcal{K}^m$, and each **row** of A is a vector in $\mathcal{K}^n$.

Definition 2.1.3 If $m = n$, the matrix A is called a **square matrix**. If $m \neq n$, A is called a **non-square matrix**. The **zero matrix** is the matrix all of whose entries are zero.

Definition 2.1.4 Let $A = (a_{ij})$ and $B = (b_{ij})$ be two matrices in $\mathbf{M}_{m\times n}(\mathcal{K})$, and let $\alpha \in \mathcal{K}$. Matrix addition and scalar multiplication are defined as:

$$(A + B) = (a_{ij} + b_{ij}), \quad (\alpha A) = (\alpha a_{ij}), \quad \text{for } 1 \leq i \leq m,\ 1 \leq j \leq n.$$

Remark 2.1.5 Let $\mathbf{M}_{m\times n}(\mathcal{K})$ (or $\mathrm{Mat}_{m\times n}(\mathcal{K})$) denote the set of all $m \times n$ matrices over $\mathcal{K}$. Equipped with the matrix addition and scalar multiplication defined above, $\mathbf{M}_{m\times n}(\mathcal{K})$ forms a **vector space** over $\mathcal{K}$.
Its dimension is

$$\dim \mathbf{M}_{m\times n}(\mathcal{K}) = mn.$$

Definition 2.1.6 The **standard basis** of the vector space $\mathbf{M}_{m\times n}(\mathcal{K})$ is the collection of mn matrices $\{e_{ij}\}$, where each e_{ij} has entry

$$(e_{ij})_{pq} = \begin{cases} 1, & \text{if } p = i \text{ and } q = j, \\ 0, & \text{otherwise.} \end{cases}$$

Each e_{ij} has a single 1 in the (i, j)-position and zeros elsewhere. Every matrix $A = (a_{ij}) \in \mathbf{M}_{m\times n}(\mathcal{K})$ can be expressed uniquely as

$$A = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} e_{ij}.$$

Definition 2.1.7 Let $A = (a_{ij}) \in \mathbf{M}_{m\times n}(\mathcal{K})$ and $B = (b_{jk}) \in \mathbf{M}_{n\times p}(\mathcal{K})$. The **product** of A and B, denoted by AB, is defined as the $m \times p$ matrix

$$AB = (c_{ik}) \in \mathbf{M}_{m\times p}(\mathcal{K}),$$

where each entry c_{ik} is given by

$$c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}, \quad \text{for } 1 \le i \le m,\ 1 \le k \le p.$$

Theorem 2.1.8 (**Associative Law**) Let $A \in \mathbf{M}_{m\times n}(\mathcal{K})$, $B \in \mathbf{M}_{n\times p}(\mathcal{K})$, and $C \in \mathbf{M}_{p\times q}(\mathcal{K})$. Then matrix multiplication is associative:

$$(AB)C = A(BC).$$

Theorem 2.1.9 (**Distributive Laws**) For matrices $A, A_1, A_2 \in \mathbf{M}_{m\times n}(\mathcal{K})$ and $B, B_1, B_2 \in \mathbf{M}_{n\times p}(\mathcal{K})$, matrix multiplication satisfies

$$A(B_1 + B_2) = AB_1 + AB_2, \quad (A_1 + A_2)B = A_1 B + A_2 B.$$

Definition 2.1.10 For each positive integer n, the **identity matrix** of order n, denoted by I_n, is the $n \times n$ matrix whose entries are defined by

$$(I_n)_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases}$$

The matrix I_n satisfies the property that for any $A \in \mathbf{M}_{m\times n}(\mathcal{K})$ and $B \in \mathbf{M}_{n\times p}(\mathcal{K})$,

$$AI_n = A, \quad I_n B = B.$$

Definition 2.1.11 Let $A \in \mathbf{M}_{n\times n}(\mathcal{K})$ be a square matrix. If there exists a matrix $B \in \mathbf{M}_{n\times n}(\mathcal{K})$ such that

$$AB = BA = I_n,$$

then A is called an **invertible matrix** (or **non-singular matrix**), and B is called the **inverse matrix** of A, denoted by A^{-1}. If no such B exists, A is called a **singular matrix**.

Remark 2.1.12 Such inverse shown above is unique.

Definition 2.1.13 Let $A = (a_{ij}) \in \mathbf{M}_{m\times n}(\mathcal{K})$. The **transpose** of A, denoted by A^T, is the $n \times m$ matrix obtained by interchanging the rows and columns of A. Formally,

$$A^T = (a_{ji}) \in \mathbf{M}_{n\times m}(\mathcal{K}),$$

that is, the (i, j)-entry of A^T equals the (j, i)-entry of A. If $A^T = A$, then A is called a **symmetric matrix**.

Theorem 2.1.14 (**Properties of the Transpose**) Let $A, B \in \mathbf{M}_{m\times n}(\mathcal{K})$ and $\alpha \in \mathcal{K}$. Then the transpose operation satisfies the following properties:

1. $(A^T)^T = A$;
2. $(A + B)^T = A^T + B^T$;
3. $(\alpha A)^T = \alpha A^T$;
4. If $A \in \mathbf{M}_{m\times n}(\mathcal{K})$ and $B \in \mathbf{M}_{n\times p}(\mathcal{K})$, then

$$(AB)^T = B^T A^T.$$

2 Linear Equations

Definition 2.2.1 A **linear system** over a field $\mathcal{K}$ is a collection of m linear equations in n variables or unknowns $x_1, x_2, \dots, x_n$ of the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \quad\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m, \end{cases}$$

where $a_{ij}, b_i \in \mathcal{K}$ for all i, j.
It can be written compactly in matrix form as

$$A\boldsymbol{x} = \boldsymbol{b},$$

where

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

Definition 2.2.2 A linear system $A\boldsymbol{x} = \boldsymbol{b}$ is called:

- **homogeneous** if $\boldsymbol{b} = \mathbf{0}$, i.e. the system is of the form $A\boldsymbol{x} = \mathbf{0}$;
- **inhomogeneous** (or nonhomogeneous) if $\boldsymbol{b} \neq \mathbf{0}$.

Definition 2.2.3 A vector $\boldsymbol{x} \in \mathcal{K}^n$ satisfying the equation $A\boldsymbol{x} = \boldsymbol{b}$ is called a **solution** of the system.
In the special case of a homogeneous system $A\boldsymbol{x} = \mathbf{0}$:

- the vector $\mathbf{0}$ is called the **trivial solution**;
- any nonzero vector $\boldsymbol{x} \neq \mathbf{0}$ satisfying $A\boldsymbol{x} = \mathbf{0}$ is called a **nontrivial solution**.

Theorem 2.2.4 Let

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0, \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0, \end{cases}$$

be a homogeneous system of m linear equations in n unknowns, with coefficients in a field $\mathcal{K}$. Assume that $n > m$. Then the system has a **nontrivial solution** in $\mathcal{K}$.

Theorem 2.2.5 Assume that $m = n$ in the system above, and that the column vectors $A^1, A^2, \ldots, A^n$ of the coefficient matrix A are linearly independent. Then the system has a **unique solution** in $\mathcal{K}$.

Chapter 3 Linear Mappings

1 Definitions

Definition 3.1.1 Let V and W be vector spaces over the same field $\mathcal{K}$. A mapping $T : V \to W$ is called a **linear mapping** (or **linear transformation**) if for all $\boldsymbol{u}, \boldsymbol{v} \in V$ and all scalars $\alpha, \beta \in \mathcal{K}$, we have

$$T(\alpha \boldsymbol{u} + \beta \boldsymbol{v}) = \alpha T(\boldsymbol{u}) + \beta T(\boldsymbol{v}).$$

Example 3.1.2 Let $A \in \mathbf{M}_{m\times n}(\mathcal{K})$. Define a mapping

$$T_A : \mathcal{K}^n \to \mathcal{K}^m, \quad T_A(\boldsymbol{x}) = A\boldsymbol{x},$$

for all $\boldsymbol{x} \in \mathcal{K}^n$. Then T_A is a **linear mapping**.
Indeed, for any $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{K}^n$ and $\alpha, \beta \in \mathcal{K}$,

$$T_A(\alpha \boldsymbol{x} + \beta \boldsymbol{y}) = A(\alpha \boldsymbol{x} + \beta \boldsymbol{y}) = \alpha A\boldsymbol{x} + \beta A\boldsymbol{y} = \alpha T_A(\boldsymbol{x}) + \beta T_A(\boldsymbol{y}).$$

Hence T_A satisfies the linearity condition, and matrix left-multiplication defines a linear mapping from $\mathcal{K}^n$ to $\mathcal{K}^m$.

Example 3.1.3 Let $V = \mathcal{K}^n$ and $W = \mathcal{K}^m$. Denote by $\mathcal{L}(V, W)$ the set of all linear mappings from V to W over the field $\mathcal{K}$. That is,

$$\mathcal{L}(V, W) = \{\, T : V \to W \mid T \text{ is linear} \,\}.$$

For $T, F \in \mathcal{L}(V, W)$ and $\alpha \in \mathcal{K}$, we define their **sum** and **scalar multiple** by

$$(T + F)(\boldsymbol{u}) = T(\boldsymbol{u}) + F(\boldsymbol{u}), \qquad (\alpha T)(\boldsymbol{u}) = \alpha\, T(\boldsymbol{u}), \quad \forall\, \boldsymbol{u} \in V.$$

With these operations of addition and scalar multiplication, $\mathcal{L}(V, W)$ forms a vector space over $\mathcal{K}$.

Definition 3.1.4 Let V be a vector space over a field $\mathcal{K}$, and let $\boldsymbol{a} \in V$. The mapping $T_{\boldsymbol{a}} : V \to V$ defined by

$$T_{\boldsymbol{a}}(\boldsymbol{x}) = \boldsymbol{x} + \boldsymbol{a},$$

for all $\boldsymbol{x} \in V$, is called a **translation** by the vector $\boldsymbol{a}$.
A translation shifts every vector in V by the same displacement $\boldsymbol{a}$. Note that a translation is generally **not** a linear mapping, since

$$T_{\boldsymbol{a}}(\mathbf{0}) = \boldsymbol{a} \neq \mathbf{0}.$$

Theorem 3.1.5 Let V and W be vector spaces over a field $\mathcal{K}$. Let $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ be a basis of V, and let $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n$ be arbitrary elements of W. Then there exists a unique linear mapping

$$T : V \to W$$

such that

$$T(\boldsymbol{v}_i) = \boldsymbol{w}_i, \quad i = 1, \ldots, n.$$

Moreover, if $x_1, \ldots, x_n \in \mathcal{K}$, then

$$T(x_1\boldsymbol{v}_1 + x_2\boldsymbol{v}_2 + \cdots + x_n\boldsymbol{v}_n) = x_1\boldsymbol{w}_1 + x_2\boldsymbol{w}_2 + \cdots + x_n\boldsymbol{w}_n.$$

2 Kernel and Image

Definition 3.2.1 Let $T : V \to W$ be a linear mapping between vector spaces over a field $\mathcal{K}$. The **kernel** of T, denoted by $\ker(T)$, is the subset of V defined by

$$\ker(T) = \{\, \boldsymbol{v} \in V \mid T(\boldsymbol{v}) = \boldsymbol{0} \,\}.$$

It consists of all vectors in V that are mapped to the zero vector in W.

Remark 3.2.2 The kernel $\ker(T)$ is a subspace of V.

Theorem 3.2.3 Let $F : V \to W$ be a linear map between vector spaces over a field $\mathcal{K}$. The following statements are equivalent:

1. $\ker(F) = \{\boldsymbol{0}\}$;
2. For all $\boldsymbol{v}, \boldsymbol{w} \in V$, if $F(\boldsymbol{v}) = F(\boldsymbol{w})$, then $\boldsymbol{v} = \boldsymbol{w}$.

In other words, F is injective if and only if its kernel is $\{\boldsymbol{0}\}$.

Remark 3.2.4 Note that F is a mapping if and only if F^{-1} is injective and surjective.

Theorem 3.2.5 Let $F : V \to W$ be a linear map whose kernel is $\{\boldsymbol{0}\}$. If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ are linearly independent elements of V, then $F(\boldsymbol{v}_1), \ldots, F(\boldsymbol{v}_n)$ are linearly independent in W.

Definition 3.2.6 Let $T : V \to W$ be a linear map between vector spaces over a field $\mathcal{K}$. The **image** of T, denoted by $\operatorname{im}(T)$, is the subset of W defined by

$$\operatorname{im}(T) = \{\, T(\boldsymbol{v}) \mid \boldsymbol{v} \in V \,\}.$$

It consists of all vectors in W that can be expressed as the image of some vector in V under T.

Remark 3.2.7 The image $\operatorname{im}(T)$ is a subspace of W.

Theorem 3.2.8 (Rank–Nullity Theorem) Let $T : V \to W$ be a linear map between finite-dimensional vector spaces over $\mathcal{K}$. Then

$$\dim(\ker T) + \dim(\operatorname{im} T) = \dim V.$$

Here, $\dim(\ker T)$ is called the **nullity** of T, and $\dim(\operatorname{im} T)$ is called the **rank** of T.

Theorem 3.2.9 (Rank–Nullity Theorem for Matrices) Let $A \in \mathbf{M}_{m\times n}(\mathcal{K})$, and let its column vectors be

$$\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots, \boldsymbol{A}_n \in \mathcal{K}^m.$$

We call

$$\operatorname{Col}(A) = \operatorname{span}\{\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n\}$$

the **column space** of A, and

$$\operatorname{Null}(A) = \{\boldsymbol{x} \in \mathcal{K}^n \mid A\boldsymbol{x} = \mathbf{0}\}$$

the **null space** of A.
Then $\operatorname{Col}(A)$ is a subspace of $\mathcal{K}^m$, and $\operatorname{Null}(A)$ is a subspace of $\mathcal{K}^n$. Moreover,

$$\dim(\operatorname{Col}(A)) + \dim(\operatorname{Null}(A)) = n,$$

where n is the number of columns of A.

Theorem 3.2.10 Let $L : V \to W$ be a linear map between finite-dimensional vector spaces over $\mathcal{K}$. Assume that

$$\dim V = \dim W.$$

If $\ker L = \{\mathbf{0}\}$, or if $\operatorname{im} L = W$, then L is **bijective**.

Definition 3.2.11 If f is a mapping, and its inverse correspondence is also a mapping, then we say f is **invertible**.

Proposition 3.2.12 A mapping f is invertible if and only if it is bijective.

Theorem 3.2.13 Suppose V and W are vector spaces over a field $\mathcal{K}$ and $T : V \to W$ is a linear map that is invertible. Then its inverse $T^{-1} : W \to V$ is also a linear map.

Theorem 3.2.14 Let V and W be vector spaces over a field $\mathcal{K}$ with

$$\dim V = \dim W,$$

Suppose a linear map $T : V \to W$, if $\ker T = \{\mathbf{0}\}$, then T is invertible.

3 Isomorphism

Definition 3.3.1 Suppose V and W are vector spaces over a field $\mathcal{K}$. Any invertible linear map $T : V \to W$ is called an **isomorphism** between V and W.

Notation 3.3.2 We say that V and W are **isomorphic** if there exists an isomorphism $T : V \to W$. In this case we denote this fact by

$$V \cong W.$$

Theorem 3.3.3 Let V and W be finite-dimensional vector spaces over a field $\mathcal{K}$. Then $V \cong W$ if and only if $\dim V = \dim W$.

Remark 3.3.4 Suppose V and W are vector spaces over a field $\mathcal{K}$ with

$$\dim V = m, \quad \dim W = n.$$

Then $V \cong \mathcal{K}^m$ and $W \cong \mathcal{K}^n$.

Theorem 3.3.5 Let U, V, W be vector spaces over a field $\mathcal{K}$, with linear maps $F : U \to V$ and $G : V \to W$. Then the composition map

$$G \circ F : U \to W$$

is also a linear map.

Chapter 4 Linear Maps and Matrices

1 The Linear Map associated with a matrix

Theorem 4.1.1 Suppose $\mathcal{L} : \mathcal{K}^n \to \mathcal{K}^m$ is a linear map. Then there exists a unique matrix $A \in \boldsymbol{M}_{m\times n}(\mathcal{K})$ such that

$$\mathcal{L}(\boldsymbol{x}) = A\boldsymbol{x}, \quad \forall \boldsymbol{x} \in \mathcal{K}^n.$$

Remark 4.1.2 Suppose $A \in \boldsymbol{M}_{m\times n}(\mathcal{K})$, we call $\mathcal{L}_A : \mathcal{K}^n \to \mathcal{K}^m$ defined by

$$\mathcal{L}_A(\boldsymbol{x}) = A\boldsymbol{x}, \quad \forall \boldsymbol{x} \in \mathcal{K}^n$$

the **linear map associated with matrix A**. From the above theorem, we also know for any linear map from $\mathcal{K}^n$ to $\mathcal{K}^m$, it is uniquely associated to a matrix.

Proposition 4.1.3 Suppose $A \in \boldsymbol{M}_{m\times n}(\mathcal{K})$ and $B \in \boldsymbol{M}_{n\times p}(\mathcal{K})$. Let $\mathcal{L}_A : \mathcal{K}^n \to \mathcal{K}^m$ and $\mathcal{L}_B : \mathcal{K}^p \to \mathcal{K}^n$ be the linear maps associated with matrices A and B respectively. Then the composition map

$$\mathcal{L}_A \circ \mathcal{L}_B : \mathcal{K}^p \to \mathcal{K}^m$$

is the linear map associated with matrix AB. We often omit the symbol $\circ$ and just write

$$\mathcal{L}_A\mathcal{L}_B = \mathcal{L}_{AB},$$

since matrix multiplication is associative.

Proposition 4.1.4 Suppose $A \in \boldsymbol{M}_{m\times n}(\mathcal{K})$ is invertible, then

$$(\mathcal{L}_A)^{-1}(\boldsymbol{x}) = \mathcal{L}_{A^{-1}}(\boldsymbol{x}), \quad \forall \boldsymbol{x} \in \mathcal{K}^n.$$

2 The Matrix Representation of a linear map

Definition 4.2.1 Suppose V and W are vector spaces over a field $\mathcal{K}$ with

$$\dim V = m, \quad \dim W = n.$$

Let $\mathcal{B}_V = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_m\}$ be a basis of V, and $\mathcal{B}_W = \{\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_n\}$ be a basis of W. We consider the following linear maps:

1. A linear map $T : V \to W$.
2. The linear map $\phi_V : V \to \mathcal{K}^m$ defined by

$$\phi_V(c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_m\boldsymbol{v}_m) = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{pmatrix} \in \mathcal{K}^m, \quad \forall c_1, c_2, \ldots, c_m \in \mathcal{K}.$$

   Note that ϕ_V is an isomorphism between V and $\mathcal{K}^m$.
3. The linear map $\phi_W : W \to \mathcal{K}^n$ defined by

$$\phi_W(d_1\boldsymbol{w}_1 + d_2\boldsymbol{w}_2 + \cdots + d_n\boldsymbol{w}_n) = \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{pmatrix} \in \mathcal{K}^n, \quad \forall d_1, d_2, \ldots, d_n \in \mathcal{K}.$$

   Note that ϕ_W is an isomorphism between W and $\mathcal{K}^n$.

Then the linear map

$$\mathcal{L} = \phi_W \circ T \circ \phi_V^{-1} : \mathcal{K}^m \to \mathcal{K}^n$$

has a unique associated matrix $A \in \boldsymbol{M}_{n\times m}(\mathcal{K})$. This matrix A is called the **matrix representation** of linear map T with respect to the bases $\mathcal{B}_V$ and $\mathcal{B}_W$, denoted by

$$A = [T]_{\mathcal{B}_V}^{\mathcal{B}_W} \quad \text{or} \quad A = \mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(T).$$

Remark 4.2.2 This is one of the ugliest notations in linear algebra.

Remark 4.2.3 Here's a graphical illustration for the matrix representation of linear maps:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, shift right, "\phi_V"'] & W \arrow[d, shift right, "\phi_W"'] \\ \mathcal{K}^m \arrow[u, shift right, "\phi_V^{-1}"'] \arrow[r, "\mathcal{L}:\mathcal{L}(x)=Ax"'] & \mathcal{K}^n \arrow[u, shift right, "\phi_W^{-1}"'] \end{tikzcd}$$

If you're not familiar with matrix representations of linear maps, remember to draw this diagram when you get confused.

Proposition 4.2.4 Let $V = W$, $\dim V = n$ and $T = \mathrm{Id}_V$ be the identity map on V with two bases $\mathcal{B}_1 = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ and $\mathcal{B}_2 = \{\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_n\}$. Then

$$\mathcal{M}_{\mathcal{B}_1}^{\mathcal{B}_2}(\mathrm{Id}_V) = \mathrm{I}_n \quad \text{if and only if} \quad \mathcal{B}_1 = \mathcal{B}_2.$$

Example 4.2.5 Let $\boldsymbol{P}_n(\mathcal{K})$ be the vector space of all polynomials with coefficients in a field $\mathcal{K}$ and of degree at most n. That is,

$$\boldsymbol{P}_n(\mathcal{K}) = \left\{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_0, a_1, \ldots, a_n \in \mathcal{K}\right\}.$$

Consider a linear map $T : \boldsymbol{P}_n(\mathcal{K}) \to \boldsymbol{P}_{n+1}(\mathcal{K})$ defined by

$$T(p) = \int_0^x p(t)\,\mathrm{d}t \quad \forall p \in \boldsymbol{P}_n(\mathcal{K}).$$

Then the matrix representation of T with respect to the standard basis $\mathcal{B}_n = \{1, x, x^2, \ldots, x^n\}$ of $\boldsymbol{P}_n(\mathcal{K})$ and the standard basis $\mathcal{B}_{n+1} = \{1, x, x^2, \ldots, x^{n+1}\}$ of $\boldsymbol{P}_{n+1}(\mathcal{K})$ is

$$\mathcal{M}_{\mathcal{B}_n}^{\mathcal{B}_{n+1}}(T) = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \cdots & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{n} & 0 \\ 0 & 0 & 0 & \cdots & 0 & \frac{1}{n+1} \end{pmatrix}.$$

Definition 4.2.6 Suppose V is a vector space over a field $\mathcal{K}$ with $\dim V = n$ and $T : V \to V$ is a linear map. If for a basis $\mathcal{B}$ of V, $\mathcal{M}_{\mathcal{B}}^{\mathcal{B}}(T)$ is a diagonal matrix, then we say the basis $\mathcal{B}$ **diagonalizes** the linear map T. If there exists such a basis $\mathcal{B}$, then we say the linear map T is **diagonalizable**. We call a matrix $A \in \boldsymbol{M}_{n\times n}(\mathcal{K})$ is **diagonalizable** if the linear map $\mathcal{L}_A : \mathcal{K}^n \to \mathcal{K}^n$ is diagonalizable.

Definition 4.2.7 Suppose $A, B \in \boldsymbol{M}_{n\times n}(\mathcal{K})$. We say A is **similar** to B if there exists an invertible matrix $P \in \boldsymbol{M}_{n\times n}(\mathcal{K})$ such that

$$A = PBP^{-1} \quad \text{or equivalently} \quad AP = PB.$$

Remark 4.2.8 Note that similarity relationship is a equivalence relationship, i.e., it is reflexive ($A \sim A$), symmetric ($A \sim B \implies B \sim A$), and transitive ($A \sim B \wedge B \sim C \implies A \sim C$). Later we will see similar matrices share lots in common.

Proposition 4.2.9 Note that $A \in \boldsymbol{M}_{n\times n}(\mathcal{K})$ is diagonalizable if and only if A is similar to a diagonal matrix. We write A in the form

$$AS = SD \text{ where } D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \text{ and } S = \begin{pmatrix} | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_n \\ | & | & & | \end{pmatrix}.$$

Hence, we have

$$A\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i, \qquad i = 1, 2, \ldots, n,$$

which motivates the following definition.

Definition 4.2.10 Suppose $A \in \boldsymbol{M}_{n\times n}(\mathcal{K})$ and $\lambda \in \mathcal{K}$ and $V \in \mathcal{K}^n$. We say λ is an **eigenvalue** of A with corresponding **eigenvector** $\boldsymbol{v} \neq \boldsymbol{0}$ if

$$A\boldsymbol{v} = \lambda\boldsymbol{v}.$$

If there exists a basis of $\mathcal{K}^n$ such that every basis vector is an eigenvector of A, we call such a basis an **eigenbasis** of A.

Theorem 4.2.11 Suppose V and W are vector spaces over a field $\mathcal{K}$ with

$$\dim V = m, \quad \dim W = n.$$

And $T : V \to W$ is a linear map. Let $\mathcal{B}_V = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_m\}$ be a basis of V, and $\mathcal{B}_W = \{\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_n\}$ be a basis of W. Let $\phi_V : V \to \mathcal{K}^m$ and $\phi_W : W \to \mathcal{K}^n$ be the isomorphisms defined as before. Then

$$\phi_W(T(\boldsymbol{v})) = \mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(T)\,\phi_V(\boldsymbol{v}), \quad \forall \boldsymbol{v} \in V.$$

Theorem 4.2.12 With the same set-up as above, suppose additionally $F : V \to W$ is another linear map. Then

$$\alpha T + \beta F : V \to W \quad \forall \alpha, \beta \in \mathcal{K}$$

is also a linear map and we have

$$\mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(\alpha T + \beta F) = \alpha \mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(T) + \beta \mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(F).$$

Thus, the mapping $\mathcal{L}(V, W) \to \boldsymbol{M}_{n\times m}(\mathcal{K})$ where $T \mapsto \mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(T)$ is a linear map.

Theorem 4.2.13 Still with the same set-up as above, additionally this time suppose $F : W \to U$ is a linear map where U is a vector space over a field $\mathcal{K}$ with $\dim U = l$ and $\mathcal{B}_U = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_l\}$ is a basis of U. We define $\phi_U : U \to \mathcal{K}^l$ accordingly. Then

$$\mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_U}(F \circ T) = \mathcal{M}_{\mathcal{B}_W}^{\mathcal{B}_U}(F)\,\mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(T).$$

Remark 4.2.14 Here's a diagram illustration for the above theorem:

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, shift right, "\phi_V"'] & W \arrow[r, "F"] \arrow[d, shift right, "\phi_W"'] & U \arrow[d, shift right, "\phi_U"'] \\
\mathcal{K}^m \arrow[r, "\mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(T)"'] \arrow[u, shift right, "\phi_V^{-1}"'] & \mathcal{K}^n \arrow[r, "\mathcal{M}_{\mathcal{B}_W}^{\mathcal{B}_U}(F)"'] \arrow[u, shift right, "\phi_W^{-1}"'] & \mathcal{K}^l \arrow[u, shift right, "\phi_U^{-1}"']
\end{tikzcd}$$

Proposition 4.2.15 Suppose V and W are isomorphic vector spaces over a field $\mathcal{K}$ with

$$\dim V = \dim W = n$$

and $T : V \to W$ is an isomorphism. Let $\mathcal{B}_V$ be a basis of V, and $\mathcal{B}_W$ be a basis of W. Then we have

$$\left(\mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_W}(T)\right)^{-1} = \mathcal{M}_{\mathcal{B}_W}^{\mathcal{B}_V}(T^{-1}).$$

Proposition 4.2.16 Assume $\mathcal{B}_1$ and $\mathcal{B}_2$ are two bases of a vector space V over a field $\mathcal{K}$ with

$$\dim V = n.$$

1. We know
$$\mathcal{M}_{\mathcal{B}_2}^{\mathcal{B}_1}(\mathrm{Id}_V)\mathcal{M}_{\mathcal{B}_1}^{\mathcal{B}_2}(\mathrm{Id}_V) = I_n = \mathcal{M}_{\mathcal{B}_1}^{\mathcal{B}_2}(\mathrm{Id}_V)\mathcal{M}_{\mathcal{B}_2}^{\mathcal{B}_1}(\mathrm{Id}_V).$$
Here these two matrices are called **change-of-basis matrices**.
2. For any linear map $T : V \to V$, we often simplify the notation $\mathcal{M}_{\mathcal{B}_V}^{\mathcal{B}_V}(T)$ to $\mathcal{M}_{\mathcal{B}_V}(T)$.
3. If $T : V \to V$ and we have two bases $\mathcal{B}_1$ and $\mathcal{B}_2$ of V, then $\mathcal{M}_{\mathcal{B}_2}(T)$ and $\mathcal{M}_{\mathcal{B}_1}(T)$ are similar matrices. In fact, we have
$$\mathcal{M}_{\mathcal{B}_2}(T) = \mathcal{M}_{\mathcal{B}_1}^{\mathcal{B}_2}(\mathrm{Id}_V)\mathcal{M}_{\mathcal{B}_1}(T)\mathcal{M}_{\mathcal{B}_2}^{\mathcal{B}_1}(\mathrm{Id}_V).$$
Using the following diagram might be more intuitive:
$$\begin{tikzcd}
\mathcal{K}^n \arrow[r, "\mathcal{M}_{\mathcal{B}_2}(T)"'] \arrow[d, shift right, "\phi_2^{-1}"'] & \mathcal{K}^n \arrow[d, shift right, "\phi_2^{-1}"'] \\
V \arrow[r, "T"] \arrow[u, shift right, "\phi_2"'] \arrow[d, shift right, "\phi_1"'] & V \arrow[u, shift right, "\phi_2"'] \arrow[d, shift right, "\phi_1"'] \\
\mathcal{K}^n \arrow[r, "\mathcal{M}_{\mathcal{B}_1}(T)"'] \arrow[u, shift right, "\phi_1^{-1}"'] & \mathcal{K}^n \arrow[u, shift right, "\phi_1^{-1}"']
\end{tikzcd}$$

Chapter 5 Scalar Products and Orthogonality

1 Scalar Products

Definition 5.1.1 Suppose V is a vector space over a field $\mathcal{K}(=\mathbb{R})$. A **scalar product** on V is a mapping

$$\langle \cdot, \cdot \rangle : V \times V \to \mathcal{K}$$

such that for all $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in V$ and all $\alpha, \beta \in \mathcal{K}$, the following properties hold:

1. (Symmetry) $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \langle \boldsymbol{v}, \boldsymbol{u} \rangle$.
2. (Linearity in the first argument) $\langle \alpha \boldsymbol{u} + \beta \boldsymbol{v}, \boldsymbol{w} \rangle = \alpha \langle \boldsymbol{u}, \boldsymbol{w} \rangle + \beta \langle \boldsymbol{v}, \boldsymbol{w} \rangle$.

Additionally, we say the scalar product is **non-degenerate** if

$$\langle \boldsymbol{v}, \boldsymbol{u} \rangle = 0, \quad \forall \boldsymbol{u} \in V \implies \boldsymbol{v} = \mathbf{0}.$$

Remark 5.1.2 The linearilty in the second argument also holds due to the conjugate symmetry.

Definition 5.1.3 Suppose V is a vector space over a field $\mathcal{K}$ with a scalar product $\langle \cdot, \cdot \rangle$. We say two vectors $\boldsymbol{u}, \boldsymbol{v} \in V$ are **orthogonal** or **perpendicular** if

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = 0.$$

Moreover, suppose $S \subseteq V$ is nonempty. We define the **orthogonal set** of S, as

$$S^{\perp} = \{\boldsymbol{v} \in V \mid \langle \boldsymbol{v}, \boldsymbol{s} \rangle = 0, \quad \forall \boldsymbol{s} \in S\}.$$

And we say $\boldsymbol{v} \in S^{\perp}$ is **orthogonal** or **perpendicular** to S and write $\boldsymbol{v} \perp S$.

Proposition 5.1.4 Suppose V is a vector space over a field $\mathcal{K}$ with a scalar product $\langle \cdot, \cdot \rangle$. If a set $S \leq V$, then $S^{\perp}$ is a subspace of V. We often call $S^{\perp}$ the **orthogonal complement** of S in V.

Example 5.1.5 Suppose $A \in \mathbf{M}_{m \times n}(\mathcal{K})$, then $\boldsymbol{x}$ is a solution of the homogeneous linear system

$$A\boldsymbol{x} = \mathbf{0}$$

if and only if $\boldsymbol{x} \in (\mathrm{Row}(A))^{\perp}$, where $\mathrm{Row}(A)$ is the row space of matrix A. That is,

$$\mathrm{Row}(A) \perp \mathrm{Null}(A).$$

Definition 5.1.6 Suppose V is a finite-dimensional vector space over a field $\mathcal{K}$ with a scalar product $\langle \cdot, \cdot \rangle$. Then V has a basis and any basis $\mathcal{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\}$ of V is called an **orthogonal basis** if

$$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0, \quad \forall i \neq j \in \{1, 2, \ldots, n\}.$$

Definition 5.1.7 Suppose V is a vector space over a field $\mathcal{K}\,(=\mathbb{R})$ with a scalar product $\langle \cdot, \cdot \rangle$. We call this scalar product **positive-definite** if

$$\langle \boldsymbol{v}, \boldsymbol{v} \rangle \begin{cases} > 0, & \boldsymbol{v} \neq \boldsymbol{0}, \\ = 0, & \boldsymbol{v} = \boldsymbol{0}. \end{cases}$$

If the scalar product is positive-definite, we define the **norm** of a vector $\boldsymbol{v} \in V$ as

$$\|\boldsymbol{v}\| = \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}.$$

Then we can define the **distance** between two vectors $\boldsymbol{u}, \boldsymbol{v} \in V$ as

$$dist(\boldsymbol{u}, \boldsymbol{v}) = \|\boldsymbol{u} - \boldsymbol{v}\|.$$

Also, if $\boldsymbol{v} \in V$ satisfies $\|\boldsymbol{v}\| = 1$, we say $\boldsymbol{v}$ is a **unit vector**. Note that for any nonzero vector $\boldsymbol{v} \in V$, the vector

$$\frac{\boldsymbol{v}}{\|\boldsymbol{v}\|}$$

is a unit vector called the **normalization** of $\boldsymbol{v}$.

Remark 5.1.8 Here we write $\mathcal{K}\,(=\mathbb{R})$ because we need "> 0" to define positive-definiteness. Actually $\mathcal{K}$ can be any field homomorphic to $\mathbb{R}$.

Example 5.1.9 Let f, g be real-valued functions integrable on the interval I containing $[0, 1]$. We define

$$\langle f, g \rangle = \int_0^1 f(x)g(x)\,\mathrm{d}x, \quad \|f\| = \sqrt{\langle f, f \rangle}.$$

Then $\langle \cdot, \cdot \rangle$ is a positive-definite scalar product on the vector space of all real-valued functions integrable on the interval I.

Proposition 5.1.10 Suppose V is a vector space over a field $\mathcal{K}\,(=\mathbb{R})$ with a positive-definite scalar product $\langle \cdot, \cdot \rangle$.

1. For $c \in \mathcal{K}$ and $\boldsymbol{v} \in V$, we have $\|c\boldsymbol{v}\| = |c|\,\|\boldsymbol{v}\|$.
2. For any $\boldsymbol{u}, \boldsymbol{v} \in V$,
$$\|\boldsymbol{u} - \boldsymbol{v}\| = \|\boldsymbol{u} + \boldsymbol{v}\| \quad \text{if and only if} \quad \boldsymbol{u} \perp \boldsymbol{v}.$$

3. (Pythagorean theorem) For any $\boldsymbol{u}, \boldsymbol{v} \in V$ with $\boldsymbol{u} \perp \boldsymbol{v}$,

$$\|\boldsymbol{u}+\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2.$$

4. (Parallelogram law) For any $\boldsymbol{u}, \boldsymbol{v} \in V$,

$$\|\boldsymbol{u}+\boldsymbol{v}\|^2 + \|\boldsymbol{u}-\boldsymbol{v}\|^2 = 2\left(\|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2\right).$$

5. (Cauchy–Schwarz inequality) For any $\boldsymbol{u}, \boldsymbol{v} \in V$,

$$|\langle \boldsymbol{u}, \boldsymbol{v}\rangle| \leq \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|.$$

6. (Triangle inequality) For any $\boldsymbol{u}, \boldsymbol{v} \in V$,

$$\|\boldsymbol{u}+\boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|.$$

7. Suppose $\boldsymbol{u}, \boldsymbol{v} \in V$ and $\boldsymbol{u} \neq \boldsymbol{0}$. Then

$$c = \frac{\langle \boldsymbol{v}, \boldsymbol{u}\rangle}{\langle \boldsymbol{u}, \boldsymbol{u}\rangle}$$

is called the **component** of $\boldsymbol{v}$ in the direction of $\boldsymbol{u}$ and we call $c\boldsymbol{u}$ the **projection** of $\boldsymbol{v}$ onto $\boldsymbol{u}$. Note that

$$(\boldsymbol{v} - c\boldsymbol{u}) \perp \boldsymbol{u}.$$

Furthermore,

$$\|\boldsymbol{v} - c\boldsymbol{u}\| = \min_{\alpha \in \mathcal{K}} \|\boldsymbol{v} - \alpha \boldsymbol{u}\|.$$

8. More generally, suppose $\boldsymbol{u_1}, \boldsymbol{u_2}, \ldots, \boldsymbol{u_k} \in V$ are non-zero vectors that are pairwise orthogonal. For any $\boldsymbol{v} \in V$, we define the **component** of $\boldsymbol{v}$ along u_i to be

$$c_i = \frac{\langle \boldsymbol{v}, \boldsymbol{u_i}\rangle}{\langle \boldsymbol{u_i}, \boldsymbol{u_i}\rangle}, \quad i = 1, 2, \ldots, k.$$

We call $c_i\boldsymbol{u_i}$ the **projection** of $\boldsymbol{v}$ onto $\boldsymbol{u_i}$. Note

$$\langle \boldsymbol{v} - \sum_{i=1}^{k} c_i \boldsymbol{u_i}, \boldsymbol{u_j}\rangle = \langle \boldsymbol{v}, \boldsymbol{u_j}\rangle - \sum_{i=1}^{k} c_i \langle \boldsymbol{u_i}, \boldsymbol{u_j}\rangle = 0, \quad j = 1, 2, \ldots, k.$$

and we have

$$\left(\boldsymbol{v} - \sum_{i=1}^{k} c_i \boldsymbol{u_i}\right) \perp \boldsymbol{u}, \quad \forall \boldsymbol{u} \in \operatorname{span}\{\boldsymbol{u_1}, \boldsymbol{u_2}, \ldots, \boldsymbol{u_k}\}.$$

We also call $\sum_{i=1}^{k} c_i \boldsymbol{u_i}$ the **projection** of $\boldsymbol{v}$ onto $\operatorname{span}\{\boldsymbol{u_1}, \boldsymbol{u_2}, \ldots, \boldsymbol{u_k}\}$. Furthermore,

$$\left\|\boldsymbol{v} - \sum_{i=1}^{k} c_i \boldsymbol{u_i}\right\| = \min_{\alpha_1, \alpha_2, \ldots, \alpha_k \in \mathcal{K}} \left\|\boldsymbol{v} - \sum_{i=1}^{k} \alpha_i \boldsymbol{u_i}\right\|.$$

Definition 5.1.11 Suppose V is a finite-dimensional vector space over a field $\mathcal{K}\,(=\mathbb{R})$ with a scalar product $\langle\cdot,\cdot\rangle$. A basis $\mathcal{B}=\{\boldsymbol{v}_1,\boldsymbol{v}_2,\ldots,\boldsymbol{v}_n\}$ of V is called an **orthonormal basis** if

$$\langle \boldsymbol{v}_i,\boldsymbol{v}_j\rangle=\begin{cases}1, & i=j,\\ 0, & i\neq j,\end{cases}\qquad \forall i,j\in\{1,2,\ldots,n\}.$$

Theorem 5.1.12 Suppose V is a finite-dimensional vector space over a field $\mathcal{K}\,(=\mathbb{R})$ with a positive-definite scalar product $\langle\cdot,\cdot\rangle$. Let $\{w_1,w_2,\ldots,w_k\}$ be an orthogonal set of vectors in V. Then for any $\boldsymbol{v}\in V$, we have

$$\langle \boldsymbol{v},\boldsymbol{v}\rangle\geq\sum_{i=1}^{k}\frac{|\langle \boldsymbol{v},w_i\rangle|^2}{\langle w_i,w_i\rangle}.$$

with equality if and only if $\boldsymbol{v}\in\operatorname{span}\{w_1,w_2,\ldots,w_k\}$.
This is called **Bessel's inequality**.

Lemma 5.1.13 Suppose V is a finite-dimensional vector space over $\mathcal{K}\,(=\mathbb{R})$ with a positive-definite scalar product $\langle\cdot,\cdot\rangle$. Let W be a proper subspace of V. If $\{w_1,w_2,\ldots,w_k\}$ is an orthonormal basis of W, then there exist vectors $w_{k+1},w_{k+2},\ldots,w_n\in V$ such that $\{w_1,w_2,\ldots,w_k,w_{k+1},w_{k+2},\ldots,w_n\}$ is an orthonormal basis of V.

Proof. This is a constructive proof via the Gram–Schmidt process.
Note that we do know there exists $x_{k+1},x_{k+2},\ldots,x_n\in V$ such that $\{w_1,w_2,\ldots,w_k,x_{k+1},x_{k+2},\ldots,x_n\}$ is a basis of V. Next we will convert this basis to an orthonormal basis via projection.
Step 1: Project x_{k+1} onto W and denote the projection as

$$w_{k+1}=x_{k+1}-\sum_{i=1}^{k}\frac{\langle x_{k+1},w_i\rangle}{\langle w_i,w_i\rangle}w_i.$$

Note that since

$$\langle w_{k+1},w_j\rangle=\langle x_{k+1},w_j\rangle-\sum_{i=1}^{k}\frac{\langle x_{k+1},w_i\rangle}{\langle w_i,w_i\rangle}\langle w_i,w_j\rangle=0,\quad j=1,2,\ldots,k,$$

we have $w_{k+1}\perp W$. Also, $w_{k+1}\neq\mathbf{0}$ since otherwise $x_{k+1}\in W$ which contradicts the fact that $\{w_1,w_2,\ldots,w_k,x_{k+1},x_{k+2},\ldots,x_n\}$ is a basis of V. Then we normalize w_{k+1}.
Inductive Step: We proceed by induction to construct the following vectors $w_{k+2},w_{k+3},\ldots,w_n$.

□

Theorem 5.1.14 Suppose V is a finite-dimensional nontrivial vector space over a field $\mathcal{K}\,(=\mathbb{R})$ with a positive-definite scalar product $\langle\cdot,\cdot\rangle$. Then V has an orthonormal basis.

Theorem 5.1.15 Suppose V is a vector space over a field $\mathcal{K}\,(=\mathbb{R})$ with a positive-definite scalar product $\langle\cdot,\cdot\rangle$ and $\dim V=n$. Define W be a nontrivial subspace of V with $\dim W=r\,(1\leq r\leq n-1)$. Define the

orthogonal complement of W in V, which is a subspace of V, as

$$W^\perp = \{\boldsymbol{v} \in V \mid \langle \boldsymbol{v}, \boldsymbol{w} \rangle = 0, \quad \forall \boldsymbol{w} \in W\}.$$

Then we have

$$\dim W + \dim W^\perp = \dim V, \quad W \oplus W^\perp = V.$$

2 Hermitian Products

For vector space V over $\mathbb{C}$, if we want to have a "scalar product" that is "postive-definite" so that

$$\langle \boldsymbol{v}, \boldsymbol{v} \rangle \in \mathbb{R} \text{ and } \langle \boldsymbol{v}, \boldsymbol{v} \rangle \geq 0,$$

we need to modify our definition of "scalar product".

Definition 5.2.1 Suppose V is a vector space over $\mathbb{C}$. A **Hermitian scalar product** on V is a mapping

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$$

satisfying the following properties:

1. (Conjugate symmetry) $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \overline{\langle \boldsymbol{v}, \boldsymbol{u} \rangle}$.
2. (Linearity in the first argument) $\langle \alpha \boldsymbol{u} + \beta \boldsymbol{v}, \boldsymbol{w} \rangle = \alpha \langle \boldsymbol{u}, \boldsymbol{w} \rangle + \beta \langle \boldsymbol{v}, \boldsymbol{w} \rangle$.

Additionally, we say the Hermitian scalar product is **positive-definite** if

$$\langle \boldsymbol{v}, \boldsymbol{v} \rangle \begin{cases} > 0, & \boldsymbol{v} \neq \boldsymbol{0}, \\ = 0, & \boldsymbol{v} = \boldsymbol{0}. \end{cases}$$

Note that

$$\langle \boldsymbol{v}, \boldsymbol{v} \rangle = \overline{\langle \boldsymbol{v}, \boldsymbol{v} \rangle} \implies \langle \boldsymbol{v}, \boldsymbol{v} \rangle \in \mathbb{R}.$$

the positive-definiteness is well-defined.

Remark 5.2.2 Note that the linearity in the second argument does not hold. Instead, we have

$$\langle \boldsymbol{w}, \alpha \boldsymbol{u} + \beta \boldsymbol{v} \rangle = \overline{\alpha} \langle \boldsymbol{w}, \boldsymbol{u} \rangle + \overline{\beta} \langle \boldsymbol{w}, \boldsymbol{v} \rangle.$$

This is called the **conjugate linearity** in the second argument.

Remark 5.2.3 For vector space V over $\mathbb{C}$, given a Hermitian product, we say x, y are orthogonal if $\langle x, y \rangle = 0\,(= \langle y, x \rangle)$. Concepts like orthogonal complement can be defined accordingly. If additionally the Hermitian product is positive-definite, concepts like norm, distance, orthogonal basis, orthonormal basis, etc. can be defined accordingly.

Definition 5.2.4 Given $A \in \mathbf{M}_{m \times n}(\mathbb{R})$, we call $\operatorname{Im}(A)$ the **column space** of A and $\operatorname{Im}(A^T)$ the **row space** of A. We also call $\dim \operatorname{Im}(A)$ the **column rank** of A and $\dim \operatorname{Im}(A^T)$ the **row rank** of A. Since the column rank and the row rank are equal, we simply call it the **rank** of A and denote it as $\text{rank}(A)$.

Remark 5.2.5 Let $A \in \mathbf{M}_{m\times n}(\mathbb{R})$ and $R = \text{rref}(A)$. Then we know $\ker(A) = \ker(R)$ and thus $\text{rank}(A) = \text{rank}(R) =$ pivots of R.

3 Bilinearity and Quadratic Form

Definition 5.3.1 Suppose U, V, W are vector spaces over a field $\mathcal{K}$ and $g : U \times V \to W$ be a map. We say g is **bilinear** if for each fixed $u \in U$, the map $v \mapsto g(u, v)$ is linear from V to W, and for each fixed $v \in V$, the map $u \mapsto g(u, v)$ is linear from U to W.

Example 5.3.2 Let $A \in \mathbf{M}_{m\times n}(\mathcal{K})$. Define $g_A : \mathcal{K}^m \times \mathcal{K}^n \to \mathcal{K}$ as $g_A(x, y) = x^T Ay$. Then g_A is a bilinear map.

Theorem 5.3.3 Given any bilinear map $g : \mathcal{K}^m \times \mathcal{K}^n \to \mathcal{K}$, there exists a unique matrix $A \in \mathbf{M}_{m\times n}(\mathcal{K})$ such that $g(x, y) = x^T Ay\,(= g_A)$ for all $x \in \mathcal{K}^m$ and $y \in \mathcal{K}^n$.

Theorem 5.3.4 The set of all bilinear maps from $\mathcal{K}^m \times \mathcal{K}^n$ to $\mathcal{K}$, denoted by $\text{Bil}(\mathcal{K}^m \times \mathcal{K}^n, \mathcal{K})$, is a vector space over $\mathcal{K}$. The association $A \mapsto g_A$ is a vector space isomorphism from $\mathbf{M}_{m\times n}(\mathcal{K})$ to $\text{Bil}(\mathcal{K}^m \times \mathcal{K}^n, \mathcal{K})$. That is, we have

$$\dim \text{Bil}(\mathcal{K}^m \times \mathcal{K}^n, \mathcal{K}) = \dim \mathbf{M}_{m\times n}(\mathcal{K}) = mn.$$

Remark 5.3.5 Note that a scalar product define on a vector space V is a bilinear map from $V \times V$ to $\mathcal{K}$. Particularly, if $V = \mathcal{K}^n$, then the unique matrix $A \in \mathbf{M}_{n\times n}(\mathcal{K})$ corresponding to the scalar product is symmetric, i.e., $A^T = A$.

Remark 5.3.6 In some textbook, a scalar product is also called a symmetric bilinear form.

Definition 5.3.7 Suppose V is a vector space over a field $\mathcal{K}$ and $\langle \cdot, \cdot \rangle$ is a scalar product on V. We call the map $f : V \to \mathcal{K}$ defined by $f(\boldsymbol{v}) = \langle \boldsymbol{v}, \boldsymbol{v} \rangle$ a **quadratic form** on V.

Example 5.3.8 For the case $V = \mathcal{K}^n$, a quadratic form on V can be uniquely associated with a symmetric matrix $A \in \mathbf{M}_{n\times n}(\mathcal{K})$ such that

$$f(x) = x^T Ax = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i a_{ij} x_j \quad \forall x \in \mathcal{K}^n.$$

Particularly, if A is a diagonal matrix, then we have

$$f(x) = \sum_{i=1}^{n} a_{ii} x_i^2.$$

4 Determinants

Recall in last semester we defined the determinant of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ as $\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$. And we conclude that A is invertible if and only if $\det(A) \neq 0$. Furthermore, we know $A^{-1} = \dfrac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ when A is invertible.

We recall more properties of determinants of 2×2 matrices:

1. $\det(I_2) = 1$.
2. Bilinearity: For any $a, b, c, d \in \mathcal{K}$, we have
$$\begin{vmatrix} a+b & c \\ d & e \end{vmatrix} = \begin{vmatrix} a & c \\ d & e \end{vmatrix} + \begin{vmatrix} b & c \\ d & e \end{vmatrix}, \quad \begin{vmatrix} a & b+c \\ d & e \end{vmatrix} = \begin{vmatrix} a & b \\ d & e \end{vmatrix} + \begin{vmatrix} a & c \\ d & e \end{vmatrix}.$$
3. Alternating property: For any $a, b, c, d \in \mathcal{K}$, we have $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = -\begin{vmatrix} c & d \\ a & b \end{vmatrix}$.
4. $\det(A) = \det\left(A^T\right)$.

Now we try to expand the definition of determinant to more general cases.

Definition 5.4.1 Based on the definition of 2×2 determinant, we define 3×3 determinant as follows: suppose
$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \in \mathbf{M}_{3\times 3}(\mathcal{K}),$$
then we define
$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Remark 5.4.2 We call this formula the **Laplace expansion** or the **cofactor formula** by the first row.

Theorem 5.4.3 The 3×3 determinant satisfies the following properties:

1. $\det(I_3) = 1$.
2. Trilinearity: as a function of the columns in $A \in \mathbf{M}_{3\times 3}(\mathcal{K})$, the determinant is linear in each column when the other two columns are fixed. For example, $\det(A_1, \alpha x + \beta y, A_3) = \alpha \det(A_1, x, A_3) + \beta \det(A_1, y, A_3)$ for any $A_1, A_3 \in \mathcal{K}^3$, $x, y \in \mathcal{K}^3$ and $\alpha, \beta \in \mathcal{K}$.
3. If two adjacent columns of A are equal, then $\det(A) = 0$.

Definition 5.4.4 Suppose function $D : \mathbf{M}_{n\times n}(\mathcal{K}) \to \mathcal{K}$ or alternatively $D : \mathcal{K}^n \times \mathcal{K}^n \times \cdots \times \mathcal{K}^n \to \mathcal{K}$ (with n copies of $\mathcal{K}^n$) with notation $D(A)$ or $D(A_1, A_2, \ldots, A_n)$ where A_i is the i-th column of A. We say:

1. D is multilinear: as a function of the columns in $A \in \mathbf{M}_{n\times n}(\mathcal{K})$, D is linear in each column when the other $n-1$ columns are fixed.
2. D is alternating: if two adjacent columns of A are equal, then $D(A) = 0$.

Theorem 5.4.5 There exists a unique multilinear and alternating function $D : \mathbf{M}_{n\times n}(\mathcal{K}) \to \mathcal{K}$ such that $D(I_n) = 1$. We call this unique D the $n \times n$ **determinant** function and denote it as $\det(\cdot)$.

Proof. We prove in a constructive and inductive way.

Existence:

Initial Cases: For $n = 1$, we define $\det([a]) = a$ for any $a \in \mathcal{K}$. For $n = 2$ and $n = 3$, we have already defined the determinant.

Inductive assumption: suppose such D exists as a map from $\mathbf{M}_{k\times k}(\mathcal{K})$ to $\mathcal{K}$ such that the three properties hold for any $k \leq n - 1$.

Inductive step: For

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix} \in \mathbf{M}_{n\times n}(\mathcal{K}),$$

we define

$$D(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} D(\tilde{A}_{ij}),$$

where $\tilde{A}_{ij}$ is the $(n-1) \times (n-1)$ matrix obtained by deleting the i-th row and the j-th column of A. By the inductive assumption, $D(\tilde{A}_{ij})$ is well-defined and satisfies the three properties. We will show that such D satisfies the three properties as well.

For multilinearity, we consider the individual term $(-1)^{i+j} a_{ij} D(\tilde{A}_{ij})$ in the sum and a particular $k \in \{1, 2, \ldots, n\}$.

- If $j \neq k$, then $(-1)^{i+j} a_{ij}$ is independent of A_k and $D(\tilde{A}_{ij})$ is linear in A_k by the inductive assumption. Thus $(-1)^{i+j} a_{ij} D(\tilde{A}_{ij})$ is linear in A_k.
- If $j = k$, then $D(\tilde{A}_{ij})$ is independent of A_k, but $(-1)^{i+j} a_{ij}$ is linear in A_k. Thus $(-1)^{i+j} a_{ij} D(\tilde{A}_{ij})$ is linear in A_k.

Hence $D(A)$ is linear in A_k.

For alternating property, suppose $A_k = A_{k+1}$ for some $k \in \{1, 2, \ldots, n-1\}$. Consider:

- If $j \neq k$ and $j \neq k+1$, then $D(\tilde{A}_{ij})$ has two adjacent columns equal and thus $D(\tilde{A}_{ij}) = 0$ by the inductive assumption. Thus $(-1)^{i+j} a_{ij} D(\tilde{A}_{ij}) = 0$.
- In the other cases, we only have

$$D(A) = (-1)^{i+k} a_{ik} D(\tilde{A}_{ik}) + (-1)^{i+k+1} a_{i,k+1} D(\tilde{A}_{i,k+1}) = 0.$$

For identity, we have

$$D(I_n) = (-1)^{i+i} D(I_{n-1}) = 1.$$

Uniqueness: This part is finished in Homework. The core idea is to utilize the permutation of columns and the alternating property to show that D is determined by its value on I_n. □

Theorem 5.4.6 Here are some additional properties of the determinant function:

1. If the i-th and j-th $(i \neq j)$ columns of A are swapped, then $\det(A)$ changes its sign.
2. If A has two identical columns or rows, then $\det(A) = 0$.
3. If one adds a scalar multiple of one column to another column, then $\det(A)$ does not change.
4. $\det(A) = \det\left(A^T\right)$.
5. $\det(AB) = \det(A)\det(B)$ for any $A, B \in \mathbf{M}_{n\times n}(\mathcal{K})$.

Theorem 5.4.7 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$ satisfies $\det(A) \neq 0$ and $b \in \mathcal{K}^n$. If $x \in \mathcal{K}^n$ is a solution to the linear system $Ax = b$, then x satisfies

$$x_i = \frac{\det(A_{ib})}{\det(A)}, \quad i = 1, 2, \ldots, n,$$

where A_{ib} is the matrix obtained by replacing the i-th column of A with b. This is called **Cramer's rule**.

Proof. Note $b = x_1 A_1 + x_2 A_2 + \cdots + x_n A_n$, where A_i is the i-th column of A. Then we have

$$\det(A_{ib}) = \det\left(A_1, \ldots, A_{i-1}, \sum_{j=1}^{n} x_j A_j, A_{i+1}, \ldots, A_n\right) = x_i \det(A).$$

$\square$

Theorem 5.4.8 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$. If $\det(A) \neq 0$, then the columns of A are linearly independent and thus A is invertible.

Proof. It's equivalent to show that if the columns of A are linearly dependent, then $\det(A) = 0$. Suppose there exists $k \in \{1, 2, \ldots, n\}$ such that

$$A_k = \sum_{j=1, j\neq k}^{n} \alpha_j A_j,$$

where $\alpha_j \in \mathcal{K}$ for $j = 1, 2, \ldots, n$ and α_j are not all zero. Then we have

$$\det(A) = \det\left(A_1, \ldots, A_{k-1}, \sum_{j=1, j\neq k}^{n} \alpha_j A_j, A_{k+1}, \ldots, A_n\right) = 0.$$

$\square$

Definition 5.4.9 Consider the following two types of column operations:

1. Add a scalar multiple of one column to another column.
2. Interchange two columns.

We call two matrices $A, B \in \mathbf{M}_{n\times n}(\mathcal{K})$ are **column equivalent** if B can be obtained from A by a succession of column operations of the above two types.

Theorem 5.4.10 Let $A, B \in \mathbf{M}_{n\times n}(\mathcal{K})$ be column equivalent. Then:

1. $\operatorname{rank}(A) = \operatorname{rank}(B)$.
2. A is invertible if and only if B is invertible.
3. $\det A = 0$ if and only if $\det B = 0$.

Proof. For (1), we only need to show that the column operations of the above two types do not change the rank of a matrix. This is straightforward since these two types of column operations do not change the linear dependence or independence of columns.
For (2), since A is invertible, we have $\operatorname{rank}(A) = n$. By (1), we have $\operatorname{rank}(B) = n$ and thus B is invertible.
For (3), since the column operations only changes the sign of the determinant, we have $\det A = 0$ if and only if $\det B = 0$. $\square$

Theorem 5.4.11 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$ is column equivalent to a lower-triangular matrix L.

Theorem 5.4.12 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$. The following conditions are equivalent:

1. A is invertible.
2. The columns of A are linearly independent.
3. $\det A \neq 0$.

Remark 5.4.13 To calculate $\det A$, we can:

1. Laplace expansion by a row or a column.
2. Permutation definition.
3. Convert A to a lower-triangular or upper-triangular matrix L by column operations of the above two types and then calculate $\det L$ as the product of diagonal entries of L.

Chapter 6 Symmetric, Hermitian and Unitary Operators

1 Operators and Transpose

Definition 6.1.1 Suppose V is a vector space over $\mathcal{K}$. A linear map $T : V \to V$ is called a **linear operator** on V.

Definition 6.1.2 Let V be a finite dimensional vector space over field $\mathcal{K}$. The **dual space** of V is denoted by V^* and is defined as

$$V^* = \mathcal{L}(V, \mathcal{K}),$$

i.e., the vector space of all linear maps from V to $\mathcal{K}$. An element of V^* is called a **linear functional** on V.

Example 6.1.3

1. Let V be a finite dimensional vector space over field $\mathcal{K}$ with a scalar product. Suppose $v_0 \in V$. Define

   $$f_{v_0} : V \to \mathcal{K}, \qquad f_{v_0}(v) = \langle v, v_0 \rangle \quad \text{for any } v \in V.$$

   Then f_{v_0} is a linear functional on V.

2. Let V be a finite dimensional vector space over $\mathbb{C}$ with a hermitian product. Suppose $v_0 \in V$. Define

   $$f_{v_0} : V \to \mathbb{C}, \qquad f_{v_0}(v) = \langle v, v_0 \rangle \quad \text{for any } v \in V.$$

   Then f_{v_0} is a linear functional on V.

   However, the map

   $$g_{v_0} : V \to \mathbb{C}, \qquad g_{v_0}(v) = \langle v_0, v \rangle \quad \text{for any } v \in V$$

   is, in general, not a linear functional on V by the conjugate linearity in the second argument of the hermitian product.

Lemma 6.1.4 Let V be a finite dimensional vector space over field $\mathcal{K}$.

1. $\dim(V) = n$ if and only if $\dim(V^*) = n$. Thus V, V^*, and $\mathcal{K}^n$ are all isomorphic to each other as vector spaces over $\mathcal{K}$.

2. Suppose $\{v_1, \ldots, v_n\}$ is a basis of V. Then there exists a unique basis $\{f_1, \ldots, f_n\}$ of V^* such that

$$f_i(v_j) = \delta_{ij} \qquad \text{for any } i, j \in \{1, \ldots, n\},$$

where δ_{ij} is the Kronecker delta function.

3. Suppose in addition that V is equipped with a non-degenerate scalar product. Then the map

$$\Phi : V \to V^*, \qquad \Phi(v) = f_v,$$

where

$$f_v(w) = \langle w, v \rangle \qquad \text{for any } w \in V,$$

is an isomorphism of vector spaces over $\mathcal{K}$.

Proof.

1. Assume $\dim(V) = n$, and let $\{v_1, \ldots, v_n\}$ be a basis of V. For each $i \in \{1, \ldots, n\}$, define

$$f_i\left(\sum_{j=1}^{n} a_j v_j\right) = a_i.$$

This is well-defined because every vector in V has a unique expression in the basis $\{v_1, \ldots, v_n\}$, and each f_i is linear.

We claim that $\{f_1, \ldots, f_n\}$ is a basis of V^*. First, it is linearly independent: if

$$c_1 f_1 + \cdots + c_n f_n = 0,$$

then evaluating at v_j gives

$$0 = (c_1 f_1 + \cdots + c_n f_n)(v_j) = c_j,$$

so all $c_j = 0$.

Next, it spans V^*. Let $f \in V^*$. Set

$$a_i = f(v_i) \qquad (i = 1, \ldots, n).$$

Then for any $v = \sum_{j=1}^{n} x_j v_j$,

$$f(v) = f\left(\sum_{j=1}^{n} x_j v_j\right) = \sum_{j=1}^{n} x_j f(v_j) = \sum_{j=1}^{n} x_j a_j = \sum_{j=1}^{n} a_j f_j(v).$$

Hence

$$f = \sum_{j=1}^{n} a_j f_j.$$

Therefore $\{f_1, \ldots, f_n\}$ is a basis of V^*, so $\dim(V^*) = n$.

Conversely, if $\dim(V^*) = n$, then applying the same result to the vector space V^* gives

$$\dim((V^*)^*) = n.$$

Since V is finite dimensional, the canonical map from V to $(V^*)^*$ is an isomorphism, hence

$$\dim(V) = \dim((V^*)^*) = n.$$

So $\dim(V) = n$ if and only if $\dim(V^*) = n$.

Since any n-dimensional vector space over $\mathcal{K}$ is isomorphic to $\mathcal{K}^n$, it follows that $V \cong V^* \cong \mathcal{K}^n$.

2. Existence has already been established in part (1): the functionals f_i defined by

$$f_i\left(\sum_{j=1}^{n} a_j v_j\right) = a_i$$

satisfy

$$f_i(v_j) = \delta_{ij}.$$

We now show uniqueness. Suppose $\{g_1, \ldots, g_n\}$ is another basis of V^* such that

$$g_i(v_j) = \delta_{ij} \qquad \text{for all } i, j.$$

Let $v = \sum_{j=1}^{n} a_j v_j$. Then

$$g_i(v) = g_i\left(\sum_{j=1}^{n} a_j v_j\right) = \sum_{j=1}^{n} a_j g_i(v_j) = \sum_{j=1}^{n} a_j \delta_{ij} = a_i = f_i(v).$$

Since this holds for every $v \in V$, we have $g_i = f_i$ for all i. Hence the basis is unique.

3. First, Φ is linear. For any $u, v, w \in V$ and $a, b \in \mathcal{K}$,

$$\Phi(au + bv) = f_{au+bv},$$

and for every $w \in V$,

$$f_{au+bv}(w) = \langle w, au + bv\rangle = a\langle w, u\rangle + b\langle w, v\rangle = af_u(w) + bf_v(w).$$

Hence

$$\Phi(au + bv) = a\Phi(u) + b\Phi(v).$$

Next, Φ is injective. If $\Phi(v) = 0$, then $f_v = 0$, i.e.

$$\langle w, v\rangle = 0 \qquad \text{for all } w \in V.$$

By non-degeneracy of the scalar product, this implies $v = 0$.

Finally, since $\dim(V) = \dim(V^*)$ by part (1), a linear injective map

$$\Phi : V \to V^*$$

between finite dimensional vector spaces of the same dimension must be surjective. Therefore Φ is an isomorphism.

$\square$

Theorem 6.1.5 Suppose V is a finite-dimensional vector space over $\mathcal{K}$ with a non-degenerate scalar product $\langle \cdot, \cdot \rangle$ and $T : V \to V$ is a linear operator. Then there exists a unique linear operator $T^T : V \to V$ such that

$$\langle T(\boldsymbol{v}), \boldsymbol{w} \rangle = \langle \boldsymbol{v}, T^T(\boldsymbol{w}) \rangle, \quad \forall \boldsymbol{v}, \boldsymbol{w} \in V.$$

We call T^T the **transpose** of T.

Proof. Let $L : V \to \mathcal{K}$ be $L(\boldsymbol{v}) = \langle T(\boldsymbol{v}), \boldsymbol{w} \rangle$ for any $\boldsymbol{v} \in V$. Note L is a linear map from V to $\mathcal{K}$ and thus a linear functional. Note by the above lemma, we know there exists a unique $\tilde{w} \in V$ such that $L(x) = f_{\tilde{w}}(x) = \langle x, \tilde{w} \rangle$. Let $T^T(\boldsymbol{w}) = \tilde{w}$.
Suppose $L_1(\boldsymbol{v}) = \langle T(\boldsymbol{v}), \boldsymbol{w}_1 \rangle$ and $\tilde{w}_1 \in V$ is the unique vector in V such that $L_1 = f_{\tilde{w}_1}$. Suppose $L_2(\boldsymbol{v}) = \langle T(\boldsymbol{v}), \boldsymbol{w}_2 \rangle$ and $\tilde{w}_2 \in V$ is the unique vector in V such that $L_2 = f_{\tilde{w}_2}$.
Note $L' := \alpha L_1 + \beta L_2$ satisfies

$$L'(\boldsymbol{v}) = \alpha \langle T(\boldsymbol{v}), \boldsymbol{w}_1 \rangle + \beta \langle T(\boldsymbol{v}), \boldsymbol{w}_2 \rangle = \langle T(\boldsymbol{v}), \alpha \boldsymbol{w}_1 + \beta \boldsymbol{w}_2 \rangle.$$

Thus $T^T(\alpha \boldsymbol{w}_1 + \beta \boldsymbol{w}_2) = \alpha T^T(\boldsymbol{w}_1) + \beta T^T(\boldsymbol{w}_2)$ and thus T^T is linear. We still need to verify $\langle T(\boldsymbol{v}), \boldsymbol{w} \rangle = \langle \boldsymbol{v}, T^T(\boldsymbol{w}) \rangle$ for all $\boldsymbol{v}, \boldsymbol{w} \in V$. This is straightforward since $L(\boldsymbol{v}) = \langle T(\boldsymbol{v}), \boldsymbol{w} \rangle = f_{\tilde{w}}(\boldsymbol{v}) = \langle \tilde{w}, \boldsymbol{v} \rangle = \langle T^T(\boldsymbol{w}), \boldsymbol{v} \rangle$. $\square$

Remark 6.1.6 We often just write $\langle T(\boldsymbol{v}), \boldsymbol{w} \rangle = \langle \boldsymbol{v}, T^T(\boldsymbol{w}) \rangle$, $\langle T\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{v}, T^T\boldsymbol{w} \rangle$ or $\langle A\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{v}, A^T\boldsymbol{w} \rangle$ where A is a general operator (instead of a matrix) on V.

Definition 6.1.7 Suppose V is a finite-dimensional vector space over field $\mathcal{K}$ and T is an operator on V. Suppose $\mathcal{B}$ is a basis for V and let $\mathcal{M}$ be the matrix representation of T with respect to basis $\mathcal{B}$, then we define the determinant of operator T to be the determinant of matrix $\mathcal{M}$, i.e., $\det(T) = \det(\mathcal{M})$.

Remark 6.1.8 Note that the determinant of an operator is well-defined since the value of the determinant of the matrix representation of T does not depend on the choice of basis $\mathcal{B}$: suppose that if a different basis $\mathcal{B}'$ is chosen and $\mathcal{M}'$ is the matrix representation of T with respect to $\mathcal{B}'$, then we have $\mathcal{M}' = P^{-1}\mathcal{M}P$ for some invertible matrix P and thus

$$\det(\mathcal{M}') = \det\left(P^{-1}\mathcal{M}P\right) = \det\left(P^{-1}\right)\det(\mathcal{M})\det(P) = \det(\mathcal{M}).$$

Theorem 6.1.9 Suppose $V, \mathcal{K}, \langle \cdot, \cdot \rangle$ satisfy the condition in Theorem 6.1.5. Let A, B be operators on V and $c \in \mathcal{K}$. Then

1. $(A+B)^T = A^T + B^T$.
2. $(cA)^T = cA^T$.
3. $(AB)^T = B^T A^T$, where AB means the composite operator: $V \to V : AB(x) = (A \circ B)(x) = A(B(x))$.
4. $(A^T)^T = A$.

2 Symmetric, Hermitian and Unitary Operators

Definition 6.2.1 Suppose $V, \mathcal{K}, \langle \cdot, \cdot \rangle$ satisfy the condition in Theorem 6.1.5 and A is an operator on V. We say A is **symmetric** if $A^T = A$.

Lemma 6.2.2 Suppose V is a finite-dimensional vector space over $\mathbb{C}$ with a positive definite hermitian product $\langle \cdot, \cdot \rangle$. Given a functional L on V, there exists a unique vector $\tilde{w} \in V$ such that $L(x) = \langle x, \tilde{w} \rangle$ for all $x \in V$.

Theorem 6.2.3 Suppose $V, \mathbb{C}, \langle \cdot, \cdot \rangle$ satisfy the condition in Lemma 6.2.2 and A is an operator on V. Then there exists a unique operator A^* on V such that $\langle Ax, y \rangle = \langle x, A^* y \rangle$ for all $x, y \in V$. We call A^* the **adjoint** or **complex transpose**, or the **conjugate transpose** of A.

Example 6.2.4 Let $V = \mathbb{C}^n$ with the standard hermitian product $\langle x, y \rangle = \sum_{i=1}^{n} x_i \overline{y_i}$. For any $A \in \mathbf{M}_{n \times n}(\mathbb{C})$, we have

$$\langle Ax, y \rangle = x^T A^T \overline{y} = x^T \overline{\left(\overline{A^T} y\right)} = \langle x, \overline{A^T} y \rangle.$$

Thus $A^* = \overline{A^T}$.

Definition 6.2.5 Suppose $V, \mathbb{C}, \langle \cdot, \cdot \rangle$ satisfy the condition in Lemma 6.2.2. An operator A on V is called **self-adjoint** or **hermitian** if $A^* = A$. Particularly, if A is hermitian, then $\langle Ax, y \rangle = \langle x, Ay \rangle$ for all $x, y \in V$.

Theorem 6.2.6 Suppose $V, \mathbb{C}, \langle \cdot, \cdot \rangle$ satisfy the condition in Lemma 6.2.2. Let A and B be operators on V and $\alpha \in \mathbb{C}$. Then

1. $(A+B)^* = A^* + B^*$.
2. $(\alpha A)^* = \overline{\alpha} A^*$.
3. $(AB)^* = B^* A^*$, where AB means the composite operator: $V \to V : AB(x) = (A \circ B)(x) = A(B(x))$.
4. $(A^*)^* = A$.

Definition 6.2.7 Suppose V is a finite-dimensional vector space over $\mathbb{R}$ with a positive definite scalar product $\langle \cdot, \cdot \rangle$. An operator A on V is called (real-)**unitary** or **orthogonal** if $\langle Ax, Ay \rangle = \langle x, y \rangle$ for all $x, y \in V$.

Theorem 6.2.8 Suppose V, $\mathbb{R}$, $\langle \cdot, \cdot \rangle$ satisfy the condition in Definition 6.2.7. Let A be an operator on V. Then the following conditions on A are equivalent:

(a) A is real-unitary, i.e., $\langle Ax, Ay\rangle = \langle x, y\rangle$ for all $x, y \in V$.

(b) A preserves norm of vectores, i.e., $\|Ax\| = \|x\|$ for all $x \in V$.

(c) If $v \in V$ is a unit vector, then Av is also a unit vector.

(d) $A^T A = I$, where I is the identity operator on V.

Proof. (a) $\Rightarrow$ (b): by definition $\|Av\| = \sqrt{\langle Av, Av\rangle} = \sqrt{\langle v, v\rangle} = \|v\|$ for any $v \in V$.
(b) $\Rightarrow$ (c): trivial.
(d) $\Rightarrow$ (a): for any $x, y \in V$, we have $\langle Ax, Ay\rangle = \langle x, A^T Ay\rangle = \langle x, y\rangle$.
(b) $\Rightarrow$ (a): for any $x, y \in V$, we have

$$\begin{aligned}\langle Ax, Ay\rangle &= \frac{1}{4}\left(\|Ax + Ay\|^2 - \|Ax - Ay\|^2\right)\\ &= \frac{1}{4}\left(\|x + y\|^2 - \|x - y\|^2\right) = \langle x, y\rangle.\end{aligned}$$

(c) $\Rightarrow$ (b):
Case 1: if $x = 0$, then $\|Ax\| = \|0\| = \|x\|$.
Case 2: if $x \neq 0$, then $\frac{x}{\|x\|}$ is a unit vector and thus

$$\left\|A\frac{x}{\|x\|}\right\|^2 = 1 = \langle \frac{x}{\|x\|}, \frac{x}{\|x\|}\rangle = \frac{1}{\|x\|^2}\langle x, x\rangle = \frac{\|x\|^2}{\|x\|^2} = 1.$$

Thus $\|Ax\| = \|x\|$.
(a) $\Rightarrow$ (d): for any $x, y \in V$, we have

$$\langle Ax, Ay\rangle = \langle x, y\rangle = \langle x, A^T Ay\rangle \implies \langle x, (A^T A - I)y\rangle = 0.$$

By the non-degeneracy of the scalar product, we have $(A^T A - I)y = 0$ for all $y \in V$ and thus $A^T A = I$. □

Example 6.2.9 Let

$$R(\theta) = \begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}.$$

For a vector $x \in \mathbb{R}^2$, Rx rotated counterclockwise by θ. We also call such a matrix unitary if the associated operator is unitary.

Definition 6.2.10 Suppose V is a finite-dimensional vector space over $\mathbb{C}$ with a positive definite hermitian product $\langle\cdot,\cdot\rangle$. An operator A on V is called (complex-)**unitary** if $\langle Ax, Ay\rangle = \langle x, y\rangle$ for all $x, y \in V$.

Theorem 6.2.11 Suppose V, $\mathbb{C}$, $\langle\cdot,\cdot\rangle$ satisfy the condition in Definition 6.2.10. Then the following conditions on an operator A on V are equivalent:

(a) A is complex-unitary, i.e., $\langle Ax, Ay\rangle = \langle x, y\rangle$ for all $x, y \in V$.

(b) A preserves norm of vectores, i.e., $\|Ax\| = \|x\|$ for all $x \in V$.

(c) If $v \in V$ is a unit vector, then Av is also a unit vector.

(d) $A^*A = I$, where I is the identity operator on V.

Proposition 6.2.12 Let V be a finite-dimensional vector space over $\mathbb{R}$ with a positive definite scalar product $\langle \cdot, \cdot \rangle$, or over $\mathbb{C}$ with a positive definite hermitian product $\langle \cdot, \cdot \rangle$. Suppose A is an operator on V. Let $\{v_1, v_2, \ldots, v_n\}$ be an orthonormal basis for V.

(a) If A is real-/complex-unitary, then $\{Av_1, Av_2, \ldots, Av_n\}$ is also an orthonormal basis for V.

(b) If $\{Av_1, Av_2, \ldots, Av_n\}$ is also an orthonormal basis for V, then A is real-/complex-unitary.

Proof. For (2), we can prove that A is unitary by showing that $\|Av\| = \|v\|$ for any $v \in V$. Since $\{v_1, v_2, \ldots, v_n\}$ is an orthonormal basis for V, we can write $v = \sum_{i=1}^{n} \alpha_i v_i$ for some unique α_i for $i = 1, 2, \ldots, n$. Then we have

$$\|v\|^2 = \langle v, v \rangle = \sum_{i=1}^{n} |\alpha_i|^2, \quad \|Av\|^2 = \langle Av, Av \rangle = \sum_{i=1}^{n} |\alpha_i|^2.$$

Thus $\|Av\| = \|v\|$ and A is unitary. $\square$

Corollary 6.2.13 Suppose V, $\mathbb{C}$, $\langle \cdot, \cdot \rangle$ satisfy the condition in Definition 6.2.10. Then the following conditions on an operator A on V are equivalent:

1. $AA^* = A^*A$.
2. $\|Av\| = \|A^*v\|$ for all $v \in V$.
3. We can write $A = S + iT$ where S and T are self-adjoint operators on V such that $ST = TS$. Moreover,

$$S = \frac{A + A^*}{2}, \quad T = \frac{A - A^*}{2i}.$$

Chapter 7 Eigenvectors and eigenvalues

1 Definitions

Recall: in Definition 4.2.10 we defined the concept of eigenvectors and eigenvalues. After introducing operators, we can discuss more properties of eigenvectors and eigenvalues in this subsection.

Definition 7.1.1 Suppose V is a vector space over $\mathcal{K}$ and A is an operator on V. We say $\lambda \in \mathcal{K}$ is an **eigenvalue** of A if there exists a nonzero vector $v \in V$ such that $Av = \lambda v$. In this case, we call v an **eigenvector** of A corresponding to eigenvalue λ.

Remark 7.1.2 Note that for a given eigenvector v of A, the corresponding eigenvalue λ is unique since $v \neq 0$. However, for a given eigenvalue λ, the corresponding eigenvectors are not unique.

Definition 7.1.3 E_λ or V_λ is defined to be a subspace of V : $E_\lambda = \{v \in V : Av = \lambda v\}$ when λ is an eigenvalue of A. We call E_λ the **eigenspace** of A corresponding to λ.

Example 7.1.4

1. Suppose V is the vector space over $\mathbb{R}$ consisting of all infinitely differentiable functions defined on $\mathbb{R}$. Let $D : V \to V$ be the differentiation operator, i.e., $D(f) = f'$. Then $e^{\lambda x}$ is an eigenvector/eigenfunction of D corresponding to eigenvalue λ for any $\lambda \in \mathbb{R}$.
2. Let
$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$
be an operator on $\mathbb{R}^2$. Then
   - if $\theta = k\pi$ for $k \in \mathbb{Z}$, then $R(\theta)$ has two eigenvalues 1 and -1.
   - if $\theta \neq k\pi$ for $k \in \mathbb{Z}$, then $R(\theta)$ has no eigenvalue.

Theorem 7.1.5 Let V be a vector space over $\mathcal{K}$ and A be an operator on V. Suppose $\{v_1, \cdots, v_m\}$ are eigenvectors of V corresponding to eigenvalues $\lambda_1, \cdots, \lambda_m$ respectively. If $\lambda_i \neq \lambda_j$ for any $i \neq j$, then $\{v_1, \cdots, v_m\}$ is linearly independent.

Proof. We can prove the theorem by induction on m. The case $m = 1$ is trivial.
Inductive step: suppose the statement holds for case m and we will prove for the case $m + 1$. Suppose

$\alpha_1, \alpha_2, \ldots, \alpha_{m+1} \in \mathcal{K}$ such that

$$\sum_{i=1}^{m+1} \alpha_i v_i = 0.$$

Applying λ_{m+1} to both sides of the above equation, we have

$$\sum_{i=1}^{m+1} \alpha_i \lambda_{m+1} v_i = 0.$$

Apllying A to both sides of the first equation, we have

$$\sum_{i=1}^{m+1} \alpha_i A v_i = \sum_{i=1}^{m+1} \alpha_i \lambda_i v_i = 0.$$

Thus

$$\sum_{i=1}^{m} \alpha_i (\lambda_{m+1} - \lambda_i) v_i = 0.$$

By the inductive hypothesis, we have $\alpha_i(\lambda_{m+1} - \lambda_i) = 0$ for $i = 1, 2, \ldots, m$. Since $\lambda_i \neq \lambda_{m+1}$ for $i = 1, 2, \ldots, m$, we have $\alpha_i = 0$ for $i = 1, 2, \ldots, m$. Thus $\alpha_{m+1} v_{m+1} = 0$ and thus $\alpha_{m+1} = 0$ since v_{m+1} is nonzero. We have shown that $\alpha_1 = \alpha_2 = \cdots = \alpha_{m+1} = 0$ and thus $\{v_1, v_2, \ldots, v_{m+1}\}$ is linearly independent. □

Proposition 7.1.6 Suppose V with $\dim V = n$ and A is an operator on V. If A has n distinct eigenvalues, then the corresponding eigenvectors form a basis $\mathcal{B}$ for V. Moreover, the matrix representation of A with respect to basis $\mathcal{B}$ is a diagonal matrix with diagonal entries being the eigenvalues of A, i.e.,

$$\mathcal{M}_{\mathcal{B}}(A) = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Remark 7.1.7 Note

$$\text{A has } n \text{ distinct eigenvalues} \implies \text{A has an eigenbasis.}$$

However,

$$\text{A has an eigenbasis} \;\not\Longrightarrow\; \text{A has } n \text{ distinct eigenvalues.}$$

For example, let $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ be an operator on $\mathbb{R}^2$. Then A has only one eigenvalue 0 but the corresponding eigenspace is $\mathbb{R}^2$ and thus any basis of $\mathbb{R}^2$ is an eigenbasis for A.

Theorem 7.1.8 Let V be a finite-dimensional vector space over $\mathcal{K}$ and A be an operator on V. Suppose $\lambda \in \mathcal{K}$. Then λ is an eigenvalue of A if and only if $A - \lambda I$ is not invertible, where I is the identity operator on V.

Definition 7.1.9 Let V be a finite-dimensional vector space over $\mathcal{K}$ and A be an operator on V. Then

$$\chi_A(t) = \det(A - tI)$$

is called the **characteristic polynomial** of A.

Definition 7.1.10 Let $\mathcal{K}$ be a field. We say $\mathcal{K}$ is **algebraically closed** if every nonconstant polynomial f in $\mathcal{K}[t]$ has a root $\lambda \in \mathcal{K}$, i.e., $f(\lambda) = 0$.

Remark 7.1.11 Note that $\mathbb{C}$ is algebraically closed but $\mathbb{R}$ is not.

Definition 7.1.12 Let V be a finite-dimensional vector space over $\mathbb{C}$ with $\dim V = n$ and A be an operator on V. Let χ_A be the characteristic polynomial of A. Note $\deg \chi_A = n$. Suppose

$$\chi_A(t) = (t-\lambda_1)^{m_1}(t-\lambda_2)^{m_2}\cdots(t-\lambda_k)^{m_k} \quad \text{with} \sum_{i=1}^{k} m_i = n$$

where $\lambda_1, \lambda_2, \ldots, \lambda_k$ are distinct eigenvalues of A. Then we call m_i the **algebraic multiplicity** of eigenvalue λ_i. We also define the **geometric multiplicity** of eigenvalue λ_i to be

$$\mu_i := \dim E_{\lambda_i}, \quad \text{where } E_{\lambda_i} = \ker(A - \lambda_i I).$$

Note the geometric multiplicity is always less than or equal to the algebraic multiplicity, i.e., $\mu_i \leq m_i$ for $i = 1, 2, \ldots, k$.

Remark 7.1.13 There're two cases that a matrix A cannot be diagonalized, i.e., A does not have an eigenbasis:

- A has no roots in $\mathbb{R}$, i.e., A has no eigenvalue. We can expand the field $\mathbb{R}$ to $\mathbb{C}$ and then A has eigenvalues in $\mathbb{C}$ since $\mathbb{C}$ is algebraically closed.
- A has eigenvalues but does not have enough linearly independent eigenvectors. For example, let $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Since the characteristic polynomial

$$\det(tI - A) = \det\begin{pmatrix} t-2 & -1 \\ 0 & t-2 \end{pmatrix} = (t-2)^2.$$

  Thus A has only one eigenvalue 2 with algebraic multiplicity 2. However, the eigenspace corresponding to eigenvalue 2 is $\{(x, 0) : x \in \mathbb{R}\}$ and thus the geometric multiplicity of eigenvalue 2 is 1. Thus A does not have an eigenbasis and thus cannot be diagonalized.

Theorem 7.1.14 If $A \in \mathbf{M}_{n\times n}(\mathbb{R})$ is symmetric, i.e., $A^T = A$, and $\lambda \in \mathbb{C}$ is an eigenvalue of A, then $\lambda \in \mathbb{R}$.

Proof. Suppose $\lambda \in \mathbb{C}$ and $x \in \mathbb{C}^n$ such that $Ax = \lambda x$. With the standard Hermitian product, we have

$$\langle Ax, x\rangle = \langle \lambda x, x\rangle = \lambda\langle x, x\rangle,$$
$$\langle Ax, x\rangle = \langle x, A^*x\rangle = \langle x, \overline{A^T}x\rangle = \langle x, \overline{A}x\rangle = \langle x, \lambda x\rangle = \overline{\lambda}\langle x, x\rangle.$$

Since $\langle x, x\rangle > 0$, we have $\lambda = \overline{\lambda}$ and thus $\lambda \in \mathbb{R}$. □

Remark 7.1.15 Note

- the eigenvectors for a symmetric real matrix are not necessarily real.
- Suppose $Ax = \lambda x$ for some $x = u + iv$ where $u, v \in \mathbb{R}^n$. Every eigenvalue λ of A must have an eigenvector in $\mathbb{R}^n$.

2 Spectral Theorems

Definition 7.2.1 Let V be a vector space over $\mathcal{K}$ and A be an operator on V. We say a subspace W of V is **invariant** under A if $Aw \in W$ for all $w \in W$. In this case, we also say A preserves W.

Example 7.2.2 Let V, A be defined as in Definition 7.2.1. Then

$$E_\lambda = \{v \in V : Av = \lambda v\} = \ker(A - \lambda I)$$

is invariant under A for any eigenvalue λ of A. This special space is called the **eigenspace** of A corresponding to eigenvalue λ.

Lemma 7.2.3 Let V be a finite-dimensional vector space over $\mathbb{R}$ with $\dim V \geq 1$ and a positive definite scalar product $\langle \cdot, \cdot \rangle$. Let A be a symmetric operator on V. If W is a subspace of V invariant under A, then the orthogonal complement $W^\perp$ of W is also invariant under A.

Proof. Suppose W is invariant under A. By definition, for any $w \in W$, $Aw \in W$. Let $v \in W^\perp$ and we will show that $Av \in W^\perp$. Since A is symmetric, we have $\langle Av, w \rangle = \langle v, Aw \rangle = 0$ for any $w \in W$. Thus $Av \in W^\perp$ and thus $W^\perp$ is invariant under A. □

Theorem 7.2.4 Let V be a finite-dimensional vector space over $\mathbb{R}$ with $\dim V \geq 1$ and a positive definite scalar product $\langle \cdot, \cdot \rangle$. If A is a symmetric operator on V, then A has a orthonormal eigenbasis.

Proof. We prove by induction on $\dim V$.
Initial case: By the real version of Theorem 7.1.14, A has a real eigenvalue λ and we pick an eigenvector with length 1 to form an orthonormal basis.
Inductive step: suppose the statement holds for V with $\dim V = n$. We need to prove for the case $\dim V = n+1$. Note that we can pick an eigenvector v_1 of A with $\|v_1\| = 1$ due to the same reasoning in initial case. Let $V_1 = \operatorname{span}\{v_1\}$ and V_1 is invariant under A. By the previous lemma, the orthogonal complement $V_1^\perp$ of V_1 is also invariant under A. Since $V_1 \oplus V_1^\perp = V$ and $\dim V_1^\perp = n$, by the inductive hypothesis, the restrictd $A : V_1^\perp \to V_1^\perp$ has an orthonormal eigenbasis $\{v_2, v_3, \ldots, v_{n+1}\}$ for $V_1^\perp$. Thus $\{v_1, v_2, \ldots, v_{n+1}\}$ is an orthonormal eigenbasis for V. □

Remark 7.2.5 A real symmetric matrix/operator is always diagonalizable with an orthonormal eigenbasis. Particularly, suppose $A \in \mathbf{M}_{n\times n}(\mathbb{R})$ is symmetric, let $\{x_1, x_2, \ldots, x_n\}$ be an orthonormal eigenbasis of A and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the corresponding eigenvalues. Then we have

$$AX = X\Lambda, \quad \text{where } X = \begin{pmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{pmatrix}, \quad \Lambda = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Note that

$$X^T X = \begin{pmatrix} \langle x_1, x_1\rangle & \langle x_1, x_2\rangle & \cdots & \langle x_1, x_n\rangle \\ \langle x_2, x_1\rangle & \langle x_2, x_2\rangle & \cdots & \langle x_2, x_n\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle x_n, x_1\rangle & \langle x_n, x_2\rangle & \cdots & \langle x_n, x_n\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = I.$$

Thus X is unitary and we call A **unitarily/orthogonally diagonalizable**. Since Λ is diagonal, we say A is **unitarily similar to a diagonal matrix** and $A = X\Lambda X^T = X\Lambda X^{-1}$ is a **spectral decomposition** of A with $\lambda_1, \lambda_2, \ldots, \lambda_n$ being the **spectrum** of A.
The converse of Theorem 7.2.4 is also true: if $A \in \mathbf{M}_{n\times n}(\mathbb{R})$ has an orthonormal eigenbasis, i.e., $A = X\Lambda X^T$ for some unitary matrix X and diagonal matrix Λ, then A is symmetric since $A^T = (X\Lambda X^T)^T = X\Lambda X^T = A$.

Theorem 7.2.6 Let V be a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and a positive definite hermitian product $\langle\cdot,\cdot\rangle$. If A is a hermitian operator on V and $\lambda \in \mathbb{C}$ is an eigenvalue of A, then $\lambda \in \mathbb{R}$.

Lemma 7.2.7 Let V and A be defined in Theorem 7.2.6. If W is a subspace of V invariant under A, then the orthogonal complement $W^\perp$ of W is also invariant under A.

Theorem 7.2.8 Let V be a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and a positive definite hermitian product $\langle\cdot,\cdot\rangle$. If A is a hermitian operator on V, then A has an orthonormal eigenbasis.

Remark 7.2.9 If $A \in \mathbf{M}_{n\times n}(\mathbb{C})$ is hermitian, then A is unitarily diagonalizable, i.e., there exists a unitary matrix X such that X^*AX is diagonal. Moreover, the converse also holds: if $A \in \mathbf{M}_{n\times n}(\mathbb{C})$ is unitarily diagonalizable, then A is hermitian.

Lemma 7.2.10 Let V be a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and a positive definite hermitian product $\langle\cdot,\cdot\rangle$. If A is a unitary operator on V and W is invariant under A, then the orthogonal complement $W^\perp$ of W is also invariant under A.

Proof. Suppose $Aw \in W$ for any $w \in W$ and A is unitary. Let $v \in W^\perp$, then $\langle v, w\rangle = 0$ for any $w \in W$. We want to show $\langle Av, w\rangle = \langle v, A^*w\rangle = \langle v, A^{-1}w\rangle = 0$ for all $w \in W$, i.e., $Av \in W^\perp$. Note that if we restrict A to be on W, i.e., $A_W : W \to W$, then $0 \in \ker A_W \leq \ker A = \{0\}$, then A_W is an isomorphism on W. Hence, $A_W^{-1} : W \to W$ is also an isomorphism and $A^{-1}w \in W$ for any $w \in W$. Thus $\langle v, A^{-1}w\rangle = 0$ for any $w \in W$ and thus $Av \in W^\perp$. □

Theorem 7.2.11 For the complex case, if A is a complex-unitary operator on V, then A has an orthonormal eigenbasis. Particularly, if $A \in \mathbf{M}_{n\times n}(\mathbb{C})$ is unitary, then $A = U\Lambda U^{-1} = U\Lambda U^*$ for some complex-unitary matrix U and diagonal matrix Λ.

Remark 7.2.12 The eigenvalues of a complex-unitary operator must satisfy $|\lambda| = 1$:

$$\langle Av, Av\rangle = \langle \lambda v, \lambda v\rangle = \lambda\overline{\lambda}\langle v, v\rangle = |\lambda|^2\langle v, v\rangle.$$

Since A is unitary, we have $\langle Av, Av\rangle = \langle v, v\rangle$ and thus $|\lambda| = 1$.

Remark 7.2.13 A complex-unitary matrix/operator is not necessarily hermitian:

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A^* = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \neq A.$$

A hermitian matrix is not necessarily complex-unitary:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad A^* = A, \quad \text{but } A^*A = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} \neq I.$$

Remark 7.2.14 Note for symmetric, hermitian and unitary operators, the spectrum theorems are quite similar, which motivates us to find a more general class of operators that can unify these three types of operators.

Definition 7.2.15 Let V be a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and a positive definite hermitian product $\langle\cdot,\cdot\rangle$. We say an operator A on V is **normal** if $A^*A = AA^*$.

Theorem 7.2.16 Let V be a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and a positive definite hermitian product $\langle\cdot,\cdot\rangle$. If A is a normal operator on V, then A has an orthonormal eigenbasis.

Remark 7.2.17 The converse still holds.

Theorem 7.2.18 For the real case, if A is a real-unitary operator on V, then V can be expressed as $V = V_1 \oplus V_2 \oplus \cdots \oplus V_r$ where

1. each V_i is a subspace of V invariant under A
2. each V_i is mutually orthogonal to each other, i.e., $\langle v_i, v_j\rangle = 0$ for any $v_i \in V_i$ and $v_j \in V_j$ with $i \neq j$
3. $\dim V_i = 1$ or 2 for $i = 1, 2, \ldots, r$.

Proof. We shall only prove the case $V = \mathbb{R}^n$ with the standard scalar product and A is a real unitary operator. The general case can be reduced to this case by choosing an orthonormal basis for V and consider the matrix representation of A with respect to this basis.
We view A as a complex-unitary matrix, i.e., $A \in \mathbf{M}_{n\times n}(\mathbb{C})$ and $A^*A = I$. Suppose $z \neq 0$, $z \in \mathbb{C}^n$ is an eigenvector of A corresponding to eigenvalue $\lambda \in \mathbb{C}$. Namely, $Az = \lambda z$ and $|\lambda| = 1$. Let $z = x + iy$ for $x, y \in \mathbb{R}^n$ and $\lambda = \cos\theta + i\sin\theta$ for some $\theta \in \mathbb{R}$. We discuss by case.
Case 1: if λ is real, then $\lambda = \pm 1$. Then $Ax + iAy = A(x+iy) = \lambda(x+iy) = \lambda x + i\lambda y$. Since $Ax, Ay, \lambda x, \lambda y$

are all real vectors, we have $Ax = \lambda x$ and $Ay = \lambda y$. We can let v to be a non-zero vector in $\{x, y\}$ and $v_1 = \dfrac{v}{\|v\|}$. We let $V_1 = \operatorname{span}\{v_1\}$, then V_1 is a one-dimensional subspace of $\mathbb{R}^n$ invariant under A. So $V_1^{\perp}$ is also invariant under A and $V = V_1 \oplus V_1^{\perp}$.

Case 2: if λ is not real, then $Az = \lambda z$ and $A\overline{z} = \overline{\lambda}\overline{z}$ since A is a real matrix. Thus $\overline{z}$ is also an eigenvector of A corresponding to eigenvalue $\overline{\lambda}$. Since $\{z, \overline{z}\}$ are linearly independent, we know $\{x, y\}$ are linearly independent. Note

$$\begin{cases} Ax = \cos\theta x - \sin\theta y \in \operatorname{span}\{x, y\}, \\ Ay = \sin\theta x + \cos\theta y \in \operatorname{span}\{x, y\}. \end{cases}$$

Let $V_1 = \operatorname{span}\{x, y\}$, then V_1 is A-invariant and $\dim V_1 = 2$. So $V_1^{\perp}$ is also invariant under A and $V = V_1 \oplus V_1^{\perp}$. By induction on n, we know V can be reduced to the direct sum of A-invariant subspaces with dimension 1 or 2. □

Corollary 7.2.19 For the real case, if A is a real-unitary operator on V, then there exists a basis $\mathcal{B}$ for V such that

$$\mathcal{M}_{\mathcal{B}}(A) = \begin{pmatrix} M_1 & 0 & \cdots & 0 \\ 0 & M_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_r \end{pmatrix}$$

where each M_i for $i = 1, 2, \ldots, r$ satisfies one of the following conditions:

$$M_i = 1, \qquad M_i = -1, \qquad M_i = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix} \text{ with } \theta_i \neq k\pi \text{ for } k \in \mathbb{Z}.$$

Remark 7.2.20 The mapping $\theta_i \mapsto -\theta_i$ gives the standard rotation matrix.

Remark 7.2.21 Operators that have orthonormal eigenbasis $\mathcal{B}$, i.e., symmetric operators, hermitian operators and complex-unitary operators, are easy to handle. However, some operators do not have orthonormal eigenbasis (or even just an eigenbasis), but it's still possible to pick a basis $\mathcal{B}$ such that $\mathcal{M}_{\mathcal{B}}(A)$ has a nice form, i.e., "is close to being diagonal".

Chapter 8 Triangulation and Jordan Form

Recall that the set of all polynomials $\mathbf{P}_n(\mathcal{K})$ with coefficients in $\mathcal{K}$ is a vector space with $\dim \mathbf{P}_n(\mathcal{K}) = n+1$. Also, the set of all polynomials $\mathcal{K}[x]$ with coefficients in $\mathcal{K}$ is a vector space with $\dim \mathcal{K}[x] = \infty$.

1 Polynomials and Triangulation of Matrices and Operators

Definition 8.1.1 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$ and $f \in \mathcal{K}[t]$ with $f(t) = a_n t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0$ for some non-negative integer n and $a_i \in \mathcal{K}$ for $i = 0, 1, \ldots, n$. We define

$$f(A) = a_n A^n + a_{n-1}A^{n-1} + \cdots + a_1 A + a_0 I_n,$$

and call it a **polynomial of matrix** A, where I_n is the identity matrix of size n.

Remark 8.1.2 We can also define $f(A)$ for any operator A on a vector space V by the same formula where $A^k := A \circ A \circ \cdots \circ A$ (k times) and I is the identity operator on V.

Theorem 8.1.3 Suppose $f, g \in \mathcal{K}[t]$ and $A \in \mathbf{M}_{n\times n}(\mathcal{K})$. Then we have

(1) for any $\alpha, \beta \in \mathcal{K}$, we have $(\alpha f + \beta g)(A) = \alpha f(A) + \beta g(A)$

(2) $(fg)(A) = f(A)g(A)$

Definition 8.1.4 A polynomial with leading coefficient 1 is called a **monic** polynomial.

Example 8.1.5 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$. Then the characteristic polynomial χ_A of A is monic.

Definition 8.1.6 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$ and $f \in \mathcal{K}[t] \setminus \{0\}$. We say that f is an **annihilating polynomial** of A if $f(A) = 0$.

Remark 8.1.7 An annihilating polynomial of A does not need to be unique.

Theorem 8.1.8 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$. Then there exists a nonzero polynomial $f \in \mathcal{K}[t]$ such that f is an annihilating polynomial of A, i.e., $f(A) = 0$.

Proof. Note $\dim \mathbf{M}_{n\times n}(\mathcal{K}) = n^2$. For any $N \geq n^2$, the set $\{I, A, A^2, \ldots, A^N\}$ is linearly dependent. Thus there exist $\alpha_0, \alpha_1, \ldots, \alpha_N \in \mathcal{K}$ such that $\alpha_0 I + \alpha_1 A + \cdots + \alpha_N A^N = 0$. Let $f(t) = \alpha_N t^N + \cdots + \alpha_1 t + \alpha_0$, then $f(A) = 0$. □

Remark 8.1.9 Theorem 8.1.3 and Theorem 8.1.8 also hold for operators on a finite-dimensional vector space V over $\mathcal{K}$.

Definition 8.1.10 Suppose V is a finite-dimensional vector space over $\mathcal{K}$ with $\dim V \geq 1$ and A is an operator on V. A set of subspaces $\{V_1, V_2, \ldots, V_n\}$ of V is called a **fan** of A in V if

(1) $V_i \subseteq V_{i+1}$ for $i = 1, 2, \ldots, n-1$,

(2) $\dim V_i = i$ for $i = 1, 2, \ldots, n$,

(3) V_i is A-invariant for $i = 1, 2, \ldots, n$, i.e., $Av \in V_i$ for any $v \in V_i$ and $i = 1, 2, \ldots, n$.

Moreover, a basis $\{v_1, \ldots, v_i\}$ of V is called a **fan basis** if $\{v_1, \ldots, v_i\}$ is a basis for V_i for $i = 1, 2, \ldots, n$.

Remark 8.1.11 Note for a given fan of A, a fan basis always exists and is not unique. However, a fan may not exist.

Theorem 8.1.12 Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a fan basis for a operator A. Then

$$\mathcal{M}_{\mathcal{B}}(A) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

is an upper-triangular matrix.

Proof. Since $V_i = \operatorname{span}\{v_1, \ldots, v_i\}$ is invariant under A for $i = 1, 2, \ldots, n$, we have $Av_i \in V_i$ for $i = 1, 2, \ldots, n$. Thus $Av_i = a_{1i}v_1 + a_{2i}v_2 + \cdots + a_{ii}v_i$ for some $a_{1i}, a_{2i}, \ldots, a_{ii} \in \mathcal{K}$ and thus $\mathcal{M}_{\mathcal{B}}(A)$ is an upper-triangular matrix. □

Remark 8.1.13 If $\mathcal{C} = \{u_1, \ldots, u_n\}$ is a basis of V such that $\mathcal{M}_{\mathcal{C}}(A)$ is an upper-triangular matrix, then $\mathcal{C}$ is a fan basis for A by letting $V_i = \operatorname{span}\{u_1, \ldots, u_i\}$ for $i = 1, 2, \ldots, n$.

Definition 8.1.14 Suppose V is a finite-dimensional vector space over $\mathcal{K}$ with $\dim V \geq 1$ and A is an operator on V. If there exists a basis $\mathcal{B}$ for V such that $\mathcal{M}_{\mathcal{B}}(A)$ is an upper-triangular matrix, then we say A is **triangulable** and $\mathcal{B}$ is a **triangulation basis** for A. Particularly, we call a sqaure matrix A **triangulable** if the associated operator is triangulable. For a triangulable matrix $A \in \mathbf{M}_{n\times n}(\mathcal{K})$, there exists an invertible matrix $S \in \mathbf{M}_{n\times n}(\mathcal{K})$ such that $S^{-1}AS$ is an upper-triangular matrix. We call the process of finding such a basis the **triangulation** of A.

Theorem 8.1.15 Suppose V is a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and A is an operator on V. Then there exists a fan of A in V and thus A is triangulable. Particularly, if $A \in \mathbf{M}_{n\times n}(\mathbb{C})$, then there exists an invertible matrix S such that $S^{-1}AS$ is an upper-triangular matrix.

Proof. We prove by induction on $n = \dim V$.
Initial case: if $n = 1$, then $\mathcal{M}_{\mathcal{B}}(A) = (a)$ for some $a \in \mathbb{C}$ for any basis $\mathcal{B}$ of V. Thus A is triangulable.
Inductive step: suppose the statement holds for $\dim V = n$. And we will prove for the case with $\dim V = n+1$. Note that we know we can pick an eigenvector $v_1 \neq 0$ of A with eigenvalue $\lambda_1 \in \mathbb{C}$. Let $V_1 = \operatorname{span}\{v_1\}$, then $V = V_1 \oplus V_1^{\perp}$.
Let

$$\begin{array}{rccc} P_1: & V & \longrightarrow & V \\ & x & \longmapsto & \operatorname{Proj}_{V_1}(x) \end{array} \qquad \begin{array}{rccc} P_2: & V & \longrightarrow & V \\ & x & \longmapsto & \operatorname{Proj}_{V_1^{\perp}}(x) \end{array}$$

Note

- if $x \in V_1$, then $P_1(x) = x \in V_1$, $P_2(x) = 0$;
- if $x \in V_1^{\perp}$, then $P_2(x) = x \in V_1^{\perp}$, $P_1(x) = 0$.

Then $V_1^{\perp}$ is P_2-invariant and P_2 can also be restricted to be on $V_1^{\perp}$ as an operator. By the inductive assumption, there exists a fan of P_2A when restricted to be from $V_1^{\perp}$ to $V_1^{\perp}$. Let $\{W_1, \dots, W_n\}$ denote such a fan and define $V_i := V_1 + W_{i-1}$ for $i = 2, 3, \dots, n+1$. Then

(1) $V_i \subseteq V_{i+1}$ for $i = 1, 2, \dots, n$ since $W_{i-1} \subseteq W_i$ for $i = 2, \dots, n+1$,

(2) $\dim V_i = \dim V_1 + \dim W_{i-1} = 1 + (i-1) = i$ because $W_{i-1} \subseteq V_1^{\perp}$

(3) V_1 is A-invariant by definition. For V_i with $i = 2, 3, \dots, n+1$, note for any $x \in V$, $(P_1 + P_2)(x) = x$. Then $A = IA = (P_1 + P_2)A = P_1A + P_2A$. It suffices to show $Av \in V_i$ for any $v \in V_i$ with $i = 2, 3, \dots, n+1$. Note $v = cv_1 + w_{i-1}$ where $c \in \mathbb{C}$ and $w_{i-1} \in W_{i-1}$. Then

$$\begin{aligned} Av &= P_1Av + P_2Av = P_1A(cv_1 + w_{i-1}) + P_2A(cv_1 + w_{i-1}) \\ &= \underbrace{(c\lambda_1)P_1v_1 + P_1Aw_{i-1}}_{\in V_1} + \underbrace{cP_2Av_1}_{=0} + \underbrace{(P_2A)w_{i-1}}_{\in W_{i-1}} \in V_1 + W_{i-1} = V_i. \end{aligned}$$

Thus V_i is A-invariant for $i = 1, 2, \dots, n+1$.

Hence, $\{V_1, V_2, \dots, V_{n+1}\}$ is a fan of A in V and thus A is triangulable. $\square$

Theorem 8.1.16 (Cayley-Hamilton Theorem) Suppose V is a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and A is an operator on V. Let

$$\chi_A(t) = \det(tI - A)$$

be the characteristic polynomial of A. Then $\chi_A(A) = 0$.

Proof. We can pick a fan $\{V_1, V_2, \dots, V_n\}$ of A in V and let $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ be a corresponding fan basis. Then we know

$$\mathcal{M}_{\mathcal{B}}(A) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

is a upper-triangular matrix. Thus

$$\chi_A(t) = \det(tI - A) = (a_{11} - t)(a_{22} - t) \cdots (a_{nn} - t)$$

and

$$\chi_A(A) = (a_{11}I - A)(a_{22}I - A)\cdots(a_{nn}I - A).$$

We will show $\prod_{j=1}^{i}(a_{jj}I - A)v = 0$ for any $v \in V_i$ by induction on i.(Note when $i = n$, we have $\chi_A(A) = 0$.)

Initial case: if $i = 1$, then for $v \in V_1 = \operatorname{span}\{v_1\}$, i.e., $v = cv_1$ for some $c \in \mathbb{C}$, we have

$$(a_{11}I - A)v = c(a_{11}I - A)v_1 = 0.$$

Inductive step: suppose the statement holds for i case, i.e., $\prod_{j=1}^{i}(a_{jj}I - A)v = 0$ for any $v \in V_i$. We will prove for $i+1$ case: for $v \in V_{i+1} = \operatorname{span}\{v_1, \ldots, v_{i+1}\} = cv_{i+1} + \widetilde{v}$ where $c \in \mathbb{C}$ and $\widetilde{v} \in \operatorname{span}\{v_1, \ldots, v_i\} = V_i$, we have

$$\prod_{j=1}^{i+1}(a_{jj}I - A)v = c\prod_{j=1}^{i}(a_{jj}I - A)\cdot(a_{i+1,i+1}I - A)v_{i+1} + \prod_{j=1}^{i}(a_{jj}I - A)\cdot(a_{i+1,i+1}I - A)\widetilde{v}.$$

Note

$$\begin{aligned}(a_{i+1,i+1}I - A)v_{i+1} &= a_{i+1,i+1}v_{i+1} - (a_{1,i+1}v_1 + a_{2,i+1}v_2 + \cdots + a_{i,i+1}v_i + a_{i+1,i+1}v_{i+1}) \\ &= -a_{1,i+1}v_1 - a_{2,i+1}v_2 - \cdots - a_{i,i+1}v_i \in V_i = \operatorname{span}\{v_1, \ldots, v_i\}.\end{aligned}$$

Thus we know the first term $\prod_{j=1}^{i}(a_{jj}I - A)\cdot(a_{i+1,i+1}I - A)v_{i+1} = 0$ by the inductive assumption. Also, note $\widetilde{v} \in V_i$ which is invariant under $(a_{i+1,i+1}I - A)$, we have $\prod_{j=1}^{i}(a_{jj}I - A)\cdot(a_{i+1,i+1}I - A)\widetilde{v} = 0$ by the inductive assumption. Hence,

$$\prod_{j=1}^{i+1}(a_{jj}I - A)v = 0 \quad \text{for any } v \in V_{i+1}.$$

By induction, we have $\prod_{j=1}^{n}(a_{jj}I - A)v = 0$ for any $v \in V_n = V$. Thus $\chi_A(A) = 0$. □

Remark 8.1.17 The Cayley-Hamilton theorem also holds for any operator on a finite-dimensional vector space over $\mathbb{R}$.

Remark 8.1.18 Suppose V is a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and A is an operator on V. So far, we know there exists a fan basis $\mathcal{B}$ such that $\mathcal{M}_{\mathcal{B}}(A)$ is upper-triangular. Later, we shall see that there exists a "particularly nice" fan basis $\widetilde{\mathcal{B}}$ such that $\mathcal{M}_{\widetilde{\mathcal{B}}}$ looks like

$$\begin{pmatrix} J_1 & 0 & \cdots & 0 \\ 0 & J_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_k \end{pmatrix}, \qquad \text{where } J_i = \begin{pmatrix} \alpha_i & 1 & 0 & \cdots & 0 \\ 0 & \alpha_i & 1 & \cdots & 0 \\ 0 & 0 & \alpha_i & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & \cdots & \alpha_i \end{pmatrix} \text{ for } i = 1, 2, \ldots, k.$$

2 Ideals in $\mathcal{K}[t]$

Theorem 8.2.1 Suppose f and g are polynomials in $\mathcal{K}[t]$. Then there exist polynomials $q, r \in \mathcal{K}[t]$ such that $f(t) = q(t)g(t) + r(t)$ with $\deg r < \deg g$. Furthermore, such q and r are unique.

Proof. Let $m = \deg g$ and $f(t) = a_n t^n + a_{n-1}t^{n-1} + \cdots + a_0$, $g(t) = b_m t^m + b_{m-1}t^{m-1} + \cdots + b_0$ with $b_m \neq 0$. We will discuss by case.
Case 1: if $n < m$, then we can let $q(t) = 0$ and $r(t) = f(t)$, then $f(t) = q(t)g(t) + r(t)$ with $\deg r < \deg g$.
Case 2: if $n \geq m$, then we prove by induction on n.
Initial case: if $n = m$, then we can let $q(t) = \dfrac{a_n}{b_m}$ and $r(t) = a_{n-1}t^{n-1} + \cdots + a_0 - q(t)(b_{m-1}t^{m-1} + \cdots + b_0)$, then $f(t) = q(t)g(t) + r(t)$ with $\deg r < \deg g$.
Inductive step: suppose the statement holds for n case, i.e., if $\deg f = n \geq m$, then there exist $q, r \in \mathcal{K}[t]$ such that $f(t) = q(t)g(t) + r(t)$ with $\deg r < \deg g$. We will prove for $n+1$ case: if $\deg f = n+1 \geq m$, then we can let $q_1(t) = \dfrac{a_{n+1}}{b_m}t^{n+1-m}$ and $r_1(t) = a_n t^n + a_{n-1}t^{n-1} + \cdots + a_0 - q_1(t)(b_{m-1}t^{m-1} + \cdots + b_0)$, then $\deg r_1 < n+1$. If $\deg r_1 < m$, then we can let $q(t) = q_1(t)$ and $r(t) = r_1(t)$, then $f(t) = q(t)g(t) + r(t)$ with $\deg r < \deg g$. If $\deg r_1 \geq m$, then by the inductive assumption, there exist $q_2, r_2 \in \mathcal{K}[t]$ such that $r_1(t) = q_2(t)g(t) + r_2(t)$ with $\deg r_2 < \deg g$. Thus we have

$$f(t) = q_1(t)g(t) + r_1(t) = (q_1(t) + q_2(t))g(t) + r_2(t).$$

Let $q(t) = q_1(t) + q_2(t)$ and $r(t) = r_2(t)$, then $f(t) = q(t)g(t) + r(t)$ with $\deg r < \deg g$.
Finally, we will show the uniqueness of q and r. Suppose there exist $q', r' \in \mathcal{K}[t]$ such that $f(t) = q'(t)g(t) + r'(t)$ with $\deg r' < \deg g$. Then we have

$$(q(t) - q'(t))g(t) = r'(t) - r(t).$$

Then $\deg((q(t) - q'(t))g(t)) \geq \deg g$ or is zero while $\deg(r'(t) - r(t)) < \deg g$ or is zero. Thus $q(t) - q'(t) = 0$ and $r'(t) - r(t) = 0$, i.e., $q(t) = q'(t)$ and $r(t) = r'(t)$. $\square$

Definition 8.2.2 We call a subset J of $\mathcal{K}[t]$ an **ideal** or a **polynomial ideal** if J satisfies the following three conditions:

(1) $0 \in J$,

(2) if $f, g \in J$, then $f + g \in J$,

(3) if $f \in J$ and $h \in \mathcal{K}[t]$, then $hf \in J$.

Remark 8.2.3 Note J being an ideal implies that J is a subspace of $\mathcal{K}[t]$. However, the converse does not hold, i.e., a subspace of $\mathcal{K}[t]$ may not be an ideal. For example, $\mathbf{P}_2(\mathcal{K})$ is a subspace of $\mathcal{K}[t]$ but is not an ideal since picking $f(t) = t^2 \in \mathbf{P}_2(\mathcal{K})$ and $h(t) = t \in \mathcal{K}[t]$, we have $h(t)f(t) = t^3 \notin \mathbf{P}_2(\mathcal{K})$.

Example 8.2.4 (1) $\{0\}$ is an ideal of $\mathcal{K}[t]$, called the **zero ideal**;

(2) $\mathcal{K}[t]$ is an ideal of $\mathcal{K}[t]$;

(3) let $f_1, f_2, \ldots, f_n \in \mathcal{K}[t]$ and

$$J := \{g \in \mathcal{K}[t] : g = g_1 f_1 + g_2 f_2 + \cdots + g_n f_n \text{ for some } g_1, g_2, \ldots, g_n \in \mathcal{K}[t]\},$$

then J is an ideal of $\mathcal{K}[t]$. We say J is **generated** by $f_1, f_2, \ldots, f_n$ and call $\{f_1, f_2, \ldots, f_n\}$ a **generating set** of J. Particularly, if a generating set of an ideal J contains only one element, we call this element a **single generator** of ideal J. Note $\{1\}$ is a single generator of $\mathcal{K}[t]$.

Remark 8.2.5 In general, an ideal may be generated by different sets; particularly, a generating set may contain several polynomials or just a single generator.

Theorem 8.2.6 Let J be an ideal of $\mathcal{K}[t]$. Then there exists a $g \in J$ such that g is a single generator of J.

Remark 8.2.7 Note, if we require the single generator to be monic, then the single generator is unique: suppose g_1 is a single generator of an ideal J and g_2 is another single generator of J. We know $g_1 = hg_2$ for some $h \in \mathcal{K}[t]$ and $\deg g_1 \geq \deg g_2$ since g_1 is a single generator of J. Similarly, we have $\deg g_2 \geq \deg g_1$. Thus $\deg g_1 = \deg g_2$ and h is a nonzero constant. Hence, g_1 and g_2 differ by a nonzero constant factor and thus if we require the single generator to be monic, then $g_1 = g_2$.

Definition 8.2.8 Suppose $f, g \in \mathcal{K}[t]$ are nonzero.

(1) We say g **devides** f (or g is a **divisor** of f) and write $g \mid f$ if there exists an $q \in \mathcal{K}[t]$ such that $f = qg$.

(2) We say $h \in \mathcal{K}[t]$ is a **common divisor** of f and g if $h \mid f$ and $h \mid g$.

(3) We say $r \in \mathcal{K}[t]$ is a **greatest common divisor** of f and g if

- r is a common divisor of f and g,
- if h is any common divisor of f and g, then $h \mid r$.

Theorem 8.2.9 Suppose f_1 and f_2 are nonzero polynomials in $\mathcal{K}[t]$. If g is a single generator for the ideal generated by f_1 and f_2, then g is a greatest common divisor of f_1 and f_2.

Remark 8.2.10 If we require the greatest common divisor to be monic, then the greatest common divisor is unique. Moreover, Theorem 8.2.9 also holds for more than two nonzero polynomials, i.e., if g is a single generator for the ideal generated by $f_1, f_2, \ldots, f_n$ with $f_i \neq 0$ for $i = 1, 2, \ldots, n$, then g is a greatest common divisor of $f_1, f_2, \ldots, f_n$.

Definition 8.2.11 Let $f_1, f_2, \ldots, f_n$ be nonzero polynomials in $\mathcal{K}[t]$. We say $f_1, f_2, \ldots, f_n$ are **relatively prime** if 1 is a common divisor of $f_1, f_2, \ldots, f_n$.

Definition 8.2.12 Suppose $A \in \mathbf{M}_{n\times n}(\mathcal{K})$. Let

$$J = \{f \in \mathcal{K}[t] : f(A) = 0\}$$

be the ideal of $\mathcal{K}[t]$ consisting of all polynomials that annihilate A. Then we call the unique monic single generator

$m_A(t)$ of J the **minimal polynomial** of A.

Remark 8.2.13 Theorem 8.2.6 guarantees the existence of the minimal polynomial $m_A(t)$. The uniqueness of the minimal polynomial $m_A(t)$ follows from Remark 8.2.7. Note $m_A(t)|\chi_A(t)$, where $\chi_A(t)$ is the characteristic polynomial of A.

3 Divisibility and Factorization of Polynomials

Corollary 8.3.1 (Bézout's Identity) Let $\mathcal{K}$ be a field and $f_1, f_2 \in \mathcal{K}[t]$. Suppose h is a greatest common divisor of f_1 and f_2. Then there exist $g_1, g_2 \in \mathcal{K}[t]$ such that

$$h = g_1 f_1 + g_2 f_2.$$

Proof. Let

$$J := \{g_1 f_1 + g_2 f_2 : g_1, g_2 \in \mathcal{K}[t]\}$$

be the ideal of $\mathcal{K}[t]$ generated by f_1 and f_2. By Theorem 8.2.9, g is a greatest common divisor of f_1 and f_2. Thus by Remark 8.2.10, there exists a nonzero constant $c \in \mathcal{K}$ such that $h = cg$. So $h \in J$ and thus by definition of J, there exist $g_1, g_2 \in \mathcal{K}[t]$ such that $h = g_1 f_1 + g_2 f_2$. □

Definition 8.3.2 Let p be a polynomial in $\mathcal{K}[t]$ with $\deg p \geq 1$. We say p is **irreducible** if given any factorization $p = fg$ with $f, g \in \mathcal{K}[t]$, then either $\deg f = 0$ or $\deg g = 0$.

Example 8.3.3 (1) if $\mathcal{K} = \mathbb{C}$, then the only irreducible polynomials in $\mathbb{C}[t]$ are the polynomials of degree 1, i.e., those of the form $c(t - \alpha)$ with $c \in \mathbb{C}$ and $\alpha \in \mathbb{C}$.

(2) if $\mathcal{K} = \mathbb{R}$, then a quadratic polynomial $at^2 + bt + c$ with $a \neq 0$ is irreducible if and only if its discriminant $b^2 - 4ac < 0$.

Lemma 8.3.4 (Euclid's Lemma) Let p be an irreducible polynomial in $\mathcal{K}[t]$. If $p \mid fg$ for some $f, g \in \mathcal{K}[t]$, then either $p \mid f$ or $p \mid g$.

Proof. Assume $p \nmid f$ and by symmetry it suffices to show $p \mid g$. Then the greatest common divisor of p and f is 1. By Bézout's identity, there exist $h_1, h_2 \in \mathcal{K}[t]$ such that $1 = h_1 p + h_2 f$. Thus

$$g = 1 \cdot g = (h_1 p + h_2 f)g = h_1 pg + h_2 fg.$$

Since $p \mid fg$, we have $fg = ph_3$ for some $h_3 \in \mathcal{K}[t]$. Thus

$$g = h_1 pg + h_2 ph_3 = p(h_1 g + h_2 h_3).$$

Hence, $p \mid g$. □

Theorem 8.3.5 (The Fundamental Theorem of Arithmetic) Every polynomial f in $\mathcal{K}[t]$ with $\deg f \geq 1$ can be factorized as a product $p_1, p_2, \ldots, p_m$ of irreducible polynomials in $\mathcal{K}[t]$. Moreover, such factorization is unique up to a nonzero constant factor and the order of the factors. Formally, if

$$f = p_1 p_2 \cdots p_m \quad \text{and} \quad f = q_1 q_2 \cdots q_n$$

are two factorizations of f with p_i and q_j irreducible for $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$, then $n = m$ and there exists nonzero constants $c_1, c_2, \ldots, c_m$ such that $p_i = c_i q_{\sigma(i)}$ for some permutation σ of $\{1, 2, \ldots, m\}$.

Proof. We prove the existence first by induction on $\deg f$.
Initial case: if $\deg f = 1$, then f is irreducible and thus the factorization of f is just f itself.
Inductive step: suppose the statement holds for any polynomial with degree $n \geq 1$. We will prove for the case with $\deg f = n + 1$. If f is irreducible, then the factorization of f is just f itself. If f is reducible, then there exist $f_1, f_2 \in \mathcal{K}[t]$ such that $f = f_1 f_2$ and $\deg f_1, \deg f_2 < \deg f = n + 1$. By the inductive assumption, there exist factorizations of f_1 and f_2 as products of irreducible polynomials. Thus we can get a factorization of f as a product of irreducible polynomials by concatenating the factorizations of f_1 and f_2.
Then, we show the uniqueness of the factorization. Suppose

$$f = p_1 p_2 \cdots p_m \quad \text{and} \quad f = q_1 q_2 \cdots q_n$$

are two factorizations of f with p_i and q_j irreducible for $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$. By Euclid's lemma, either $p_1 \mid q_1$ or $p_1 \mid q_j$ for some $j = 2, 3, \ldots, n$. Now we rearrange the order of $q_1, q_2, \ldots, q_n$ such that $p_1 \mid q_1$. Since p_1 and q_1 are irreducible, we have $p_1 = c_1 q_1$ for some nonzero constant c_1. Thus

$$c_1 q_1 p_2 \cdots p_m = p_1 p_2 \cdots p_m = f = q_1 q_2 \cdots q_n.$$

Canceling q_1 on both sides, we have

$$c_1 p_2 \cdots p_m = q_2 \cdots q_n.$$

By repeating the above process, we can get $p_i = c_i q_i$ for some nonzero constant c_i and $i = 1, 2, \ldots, m$. By symmetry, we have $n = m$ and there exists nonzero constants $c_1, c_2, \ldots, c_m$ such that $p_i = c_i q_i$ for $i = 1, 2, \ldots, m$. □

Corollary 8.3.6 Let $f \in \mathcal{K}[t]$ with $\deg f \geq 1$. Then there exist irreducible monic polynomials $p_1, p_2, \ldots, p_m$ in $\mathcal{K}[t]$ such that

$$f = c p_1 p_2 \cdots p_m \quad \text{for some } c \in \mathcal{K}, c \neq 0.$$

Moreover, such factorization is unique up to the order of the factors.

Corollary 8.3.7 Let $f \in \mathbb{C}[t]$ with $\deg f = n \geq 1$. Then there exist $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{C}$ such that

$$f = c(t - \alpha_1)(t - \alpha_2) \cdots (t - \alpha_n) \quad \text{for some } c \in \mathbb{C}, c \neq 0.$$

Definition 8.3.8 Suppose f and g are nonzero polynomials in $\mathcal{K}[t]$ and $p \in \mathcal{K}[t]$ is irreducible. If $f = p^m g$ for some $m \geq 0$ and $p \nmid g$, then we call m the **multiplicity** of p in f. Moreover, if $p = t - \alpha$ for some $\alpha \in \mathbb{C}$, then we call m the **multiplicity** of α in f.

4 Cyclic Decomposition and Jordan Form

Proposition 8.4.1 Let $f \in \mathcal{K}[t]$ and $W = \ker f(A)$. Then W is an A-invariant subspace of V.

Proof. Let $\boldsymbol{w} \in W$, i.e., $f(A)\boldsymbol{w} = \mathbf{0}$. Then by the commutativity of polynomials of matrices, we have

$$[f(A)](A\boldsymbol{w}) = [f(A)A]\boldsymbol{w} = [Af(A)]\boldsymbol{w} = A[f(A)\boldsymbol{w}] = A\mathbf{0} = \mathbf{0}.$$

Thus $A\boldsymbol{w} \in W$. Hence, W is an A-invariant subspace of V. □

Theorem 8.4.2 Let $f \in \mathcal{K}[t]$ such that $f = f_1 f_2$ for some $f_1, f_2 \in \mathcal{K}[t]$ where the greatest common divisor of f_1 and f_2 is 1. Suppose V is a vector space and A is an operator on V such that $f(A) = 0$. Then we have

$$V = \ker f_1(A) \oplus \ker f_2(A).$$

Proof. By Bézout's identity, there exist $g_1, g_2 \in \mathcal{K}[t]$ such that

$$1 = g_1 f_1 + g_2 f_2.$$

Hence we have

$$I = g_1(A)f_1(A) + g_2(A)f_2(A).$$

Let $\boldsymbol{v} \in V$. Then applying both sides to $\boldsymbol{v}$, we have

$$\boldsymbol{v} = \underbrace{g_1(A)f_1(A)\boldsymbol{v}}_{\in \ker f_2(A)} + \underbrace{g_2(A)f_2(A)\boldsymbol{v}}_{\in \ker f_1(A)}.$$

Since

$$f_2(A)[g_1(A)f_1(A)\boldsymbol{v}] = g_1(A)[f_1(A)f_2(A)]\boldsymbol{v} = g_1(A)f(A)\boldsymbol{v} = \mathbf{0},$$

we have $g_1(A)f_1(A)\boldsymbol{v} \in \ker f_2(A)$. Similarly, we have $g_2(A)f_2(A)\boldsymbol{v} \in \ker f_1(A)$. Thus $V = \ker f_1(A) + \ker f_2(A)$.

Then it suffices to show $\ker f_1(A) \cap \ker f_2(A) = \{\mathbf{0}\}$. Let $\boldsymbol{w} \in \ker f_1(A) \cap \ker f_2(A)$, i.e., $f_1(A)\boldsymbol{w} = f_2(A)\boldsymbol{w} = \mathbf{0}$. Then

$$\boldsymbol{w} = I\boldsymbol{w} = [g_1(A)f_1(A) + g_2(A)f_2(A)]\boldsymbol{w} = g_1(A)f_1(A)\boldsymbol{w} + g_2(A)f_2(A)\boldsymbol{w} = \mathbf{0}.$$

Hence, $\ker f_1(A) \cap \ker f_2(A) = \{\mathbf{0}\}$ and thus $V = \ker f_1(A) \oplus \ker f_2(A)$. □

Theorem 8.4.3 Let V be a vector space over $\mathbb{C}$ and A be an operator on V. Let $P \in \mathbb{C}[t]$ be a monic polynomial such that $P(A) = 0$. If we write P as

$$P = (t - \alpha_1)^{m_1}(t - \alpha_2)^{m_2} \cdots (t - \alpha_k)^{m_k}$$

where $\alpha_i \in \mathbb{C}$ are the distinct roots and $m_i \geq 1$ are the multiplicities of α_i for $i = 1, 2, \ldots, k$, then we have

$$V = \bigoplus_{i=1}^{k} \ker(A - \alpha_i I)^{m_i}.$$

Proof. We prove the theorem by induction on k, the number of distinct roots of P.
Base case. If $k = 1$, then $P(t) = (t - \alpha_1)^{m_1}$. Since $P(A) = 0$, we have $(A - \alpha_1 I)^{m_1} = 0$, so every vector in V lies in $\ker(A - \alpha_1 I)^{m_1}$. Hence

$$V = \ker(A - \alpha_1 I)^{m_1}.$$

Inductive step. Suppose the result holds for polynomials with $k - 1$ distinct roots. Now let

$$P(t) = \prod_{i=1}^{k} (t - \alpha_i)^{m_i}$$

and assume $P(A) = 0$. Write $g_i(t) = (t - \alpha_i)^{m_i}$. Since g_1 and $g_2 \cdots g_k$ are relatively coprime, Theorem 8.4.2 gives

$$V = \ker g_1(A) \oplus W, \qquad W = \ker\big(g_2(A) \cdots g_k(A)\big).$$

Let

$$Q(t) = \prod_{i=2}^{k} (t - \alpha_i)^{m_i}, \quad W = \ker Q(A)$$

Since $Q(A)$ commutes with A, Proposition 8.4.1 implies that W is A-invariant. Thus A restricts to an operator $B = A|_W$ on W. By the definition of W, we have $Q(A)w = 0$ for every $w \in W$. Since $B = A|_W$, this means $Q(B) = Q(A)|_W = 0$.
The polynomial Q has $k - 1$ distinct roots, namely $\alpha_2, \ldots, \alpha_k$. By the inductive hypothesis,

$$W = \bigoplus_{i=2}^{k} \ker(B - \alpha_i I_W)^{m_i}.$$

We now identify these kernels with the corresponding kernels for A. Since $B = A|_W$,

$$\ker(B - \alpha_i I_W)^{m_i} = W \cap \ker(A - \alpha_i I_V)^{m_i}.$$

It remains to show that $\ker(A - \alpha_i I_V)^{m_i} \subseteq W$ for $i = 2, \ldots, k$. If $v \in \ker(A - \alpha_i I_V)^{m_i}$, then $g_i(A)v = 0$. Since polynomials in A commute,

$$Q(A)v = \left(\prod_{j=2}^{k} g_j(A) \right) v = \left(\prod_{\substack{j=2 \\ j \neq i}}^{k} g_j(A) \right) g_i(A)v = 0.$$

Thus $v \in \ker Q(A) = W$. Therefore

$$\ker(B - \alpha_i I_W)^{m_i} = \ker(A - \alpha_i I_V)^{m_i}, \quad \text{for } i = 2, \ldots, k.$$

Combining the decompositions, we get

$$V = \ker g_1(A) \oplus W = \ker(A - \alpha_1 I)^{m_1} \oplus \bigoplus_{i=2}^{k} \ker(A - \alpha_i I)^{m_i} = \bigoplus_{i=1}^{k} \ker(A - \alpha_i I)^{m_i}.$$

This completes the proof. □

Definition 8.4.4 Let V be a vector space over a field $\mathcal{K}$ and $\mathcal{S}$ be a set of operators on V. We say a subspace W of V is **$\mathcal{S}$-invariant** if W is invariant under every operator in $\mathcal{S}$, i.e., for any $A \in \mathcal{S}$ and $\boldsymbol{w} \in W$, we have $A\boldsymbol{w} \in W$. Moreover, we say V is a **simple $\mathcal{S}$-space** if $V \neq \{\mathbf{0}\}$ and the only $\mathcal{S}$-invariant subspaces of V are $\{\mathbf{0}\}$ and V itself.

Proposition 8.4.5 Let V be a vector space over a field $\mathcal{K}$ and $\mathcal{S}$ be a set of operators on V. Suppose A is an operator on V such that

$$AB = BA \quad \text{for any } B \in \mathcal{S}.$$

Then $\ker A$ and $\operatorname{im} A$ are both $\mathcal{S}$-invariant subspaces of V.

Proof. Let $\boldsymbol{w} \in \ker A$, i.e., $A\boldsymbol{w} = \mathbf{0}$. Then for any $B \in \mathcal{S}$, we have

$$A(B\boldsymbol{w}) = B(A\boldsymbol{w}) = B\mathbf{0} = \mathbf{0}.$$

Thus $B\boldsymbol{w} \in \ker A$. Hence, $\ker A$ is an $\mathcal{S}$-invariant subspace of V.
Let $\boldsymbol{v} \in \operatorname{im} A$, i.e., there exists $\widetilde{\boldsymbol{v}} \in V$ such that $A\widetilde{\boldsymbol{v}} = \boldsymbol{v}$. Then for any $B \in \mathcal{S}$, we have

$$B\boldsymbol{v} = B(A\widetilde{\boldsymbol{v}}) = A(B\widetilde{\boldsymbol{v}}).$$

Thus $B\boldsymbol{v} \in \operatorname{im} A$. Hence, $\operatorname{im} A$ is an $\mathcal{S}$-invariant subspace of V. $\square$

Proposition 8.4.6 Let V be a vector space over a field $\mathcal{K}$ and $\mathcal{S}$ be a set of operators on V. Suppose A is an operator on V such that

$$AB = BA \quad \text{for any } B \in \mathcal{S}.$$

If f is a polynomial in $\mathcal{K}[t]$, then

$$f(A)B = Bf(A) \quad \text{for any } B \in \mathcal{S}.$$

Proof. Let $f(t) = a_n t^n + a_{n-1}t^{n-1} + \cdots + a_0$ with $a_n \neq 0$. Then for any $B \in \mathcal{S}$, we have

$$f(A)B = (a_n A^n + a_{n-1}A^{n-1} + \cdots + a_0 I)B = a_n A^n B + a_{n-1}A^{n-1}B + \cdots + a_0 B.$$

Since $AB = BA$ for any $B \in \mathcal{S}$, we have

$$A^2B = A(AB) = A(BA) = (AB)A = (BA)A = BA^2,$$

and by repeating the above process, we can get $A^kB = BA^k$ for any positive integer k. Thus

$$f(A)B = a_n BA^n + a_{n-1}BA^{n-1} + \cdots + a_0 B = Bf(A).$$

$\square$

Theorem 8.4.7 (Schur's Lemma) Let V be a vector space over a field $\mathcal{K}$ and $\mathcal{S}$ be a set of operators on V. Suppose V is a simple $\mathcal{S}$-space and A is an operator on V such that

$$AB = BA \quad \text{for any } B \in \mathcal{S}.$$

Then either A is invertible or $A = 0$.

Proof. Since V is simple, $V \neq \{\mathbf{0}\}$ and A cannot be both invertible and the zero operator. Moreover, if $A = 0$, then the statement trivially holds. Thus we assume $A \neq 0$ and we will show A is invertible.
By Proposition 8.4.5, $\ker A$ and $\operatorname{im} A$ are both $\mathcal{S}$-invariant subspaces of V. Since V is simple, its only $\mathcal{S}$-invariant subspaces are $\{\mathbf{0}\}$ and V itself. Thus

$$\begin{cases} \ker A = \{\mathbf{0}\} \\ \operatorname{im} A = V \end{cases} \quad \text{or} \quad \begin{cases} \ker A = V \\ \operatorname{im} A = \{\mathbf{0}\} \end{cases}$$

Since $A \neq 0$, we cannot have $\ker A = V$ and nor can we have $\operatorname{im} A = \{\mathbf{0}\}$. Thus $\ker A = \{\mathbf{0}\}$ and $\operatorname{im} A = V$. Hence, A is injective and surjective, i.e., A is invertible. $\square$

Theorem 8.4.8 Let V be a finite-dimensional vector space over $\mathbb{C}$ and $\mathcal{S}$ be a set of operators on V. Suppose V is a simple $\mathcal{S}$-space and A is an operator on V such that

$$AB = BA \quad \text{for any } B \in \mathcal{S}.$$

Then there exists $\lambda \in \mathbb{C}$ such that $A = \lambda I$.

Proof. Let g be the minimal polynomial of A and we will prove g to be irreducible. Suppose, for a contradiction, g is reducible in $\mathbb{C}[t]$. Then there exist nonconstant polynomials $g_1, g_2 \in \mathbb{C}[t]$ such that $g = g_1 g_2$. Note g is the minimal polynomial of A, so $g_1(A) \neq 0$ and $g_2(A) \neq 0$. By Theorem 8.4.7, $g_1(A)$ and $g_2(A)$ are both invertible. Since $g(A) = g_1(A)g_2(A) = 0$, we get a contradiction. Hence, g is irreducible in $\mathbb{C}[t]$. Note $\mathbb{C}$ is algebraically closed. Hence, the only irreducible polynomials in $\mathbb{C}[t]$ are of the form $t - \lambda$ for some $\lambda \in \mathbb{C}$, and we have $g(t) = t - \lambda$ for some $\lambda \in \mathbb{C}$. Thus $A = g(A) + \lambda I = \lambda I$. $\square$

Definition 8.4.9 Suppose V is a finite-dimensional vector space over $\mathcal{K}$ and A is an operator on V. We say A is **nilpotent** if there exists a positive integer $k \geq 1$ such that $A^k = 0$.

Definition 8.4.10 Let V be a vector space over $\mathbb{C}$ and A be an operator on V. Suppose $\boldsymbol{v} \in V$ is a nonzero vector. We say $\boldsymbol{v}$ is **$(A - \alpha I)$-cyclic** if there exists a positive integer r such that

$$(A - \alpha I)^r \boldsymbol{v} = \mathbf{0}, \quad \alpha \in \mathbb{C}.$$

The smallest positive integer r satisfying the above condition is called the **period** of $\boldsymbol{v}$.

Lemma 8.4.11 Let V be a vector space over $\mathbb{C}$ and A be an operator on V. Suppose $\boldsymbol{v} \in V$ is an $(A - \alpha I)$-cyclic vector with period r and $\alpha \in \mathbb{C}$. Then the vectors

$$\boldsymbol{v}, (A - \alpha I)\boldsymbol{v}, (A - \alpha I)^2 \boldsymbol{v}, \ldots, (A - \alpha I)^{r-1} \boldsymbol{v}$$

are linearly independent.

Proof. For convenience, we write $B := A - \alpha I$. Suppose there exist $c_0, c_1, \ldots, c_{r-1} \in \mathbb{C}$ such that

$$c_0\boldsymbol{v} + c_1 B\boldsymbol{v} + c_2 B^2\boldsymbol{v} + \cdots + c_{r-1}B^{r-1}\boldsymbol{v} = \mathbf{0}.$$

Let k be the smallest nonnegative integer such that $c_k \neq 0$. Then we have

$$c_k B^k\boldsymbol{v} + c_{k+1}B^{k+1}\boldsymbol{v} + \cdots + c_{r-1}B^{r-1}\boldsymbol{v} = \mathbf{0}.$$

Applying B^{r-1-k} to both sides, we have

$$c_k B^{r-1}\boldsymbol{v} = \mathbf{0}.$$

Since $c_k \neq 0$ and $B^{r-1}\boldsymbol{v} \neq \mathbf{0}$, we get a contradiction. Hence, such k does not exist, and the vectors $\boldsymbol{v}, B\boldsymbol{v}, B^2\boldsymbol{v}, \ldots, B^{r-1}\boldsymbol{v}$ are linearly independent. □

Definition 8.4.12 Let V be a finite-dimensional vector space over $\mathbb{C}$ and A be an operator on V. We say V is **cyclic with respect to** A if there exists a nonzero vector $\boldsymbol{v} \in V$ such that $\boldsymbol{v}$ is $(A - \alpha I)$-cyclic with period r for some $\alpha \in \mathbb{C}$, and the ordered set

$$\mathcal{B} := \{(A - \alpha I)^{r-1}\boldsymbol{v}, (A - \alpha I)^{r-2}\boldsymbol{v}, \ldots, (A - \alpha I)\boldsymbol{v}, \boldsymbol{v}\}$$

is a basis of V. In this case, we call $\mathcal{B}$ a **Jordan basis** for V with respect to A. Note

$$A(A - \alpha I)^j\boldsymbol{v} = (A - \alpha I)^{j+1}\boldsymbol{v} + \alpha(A - \alpha I)^j\boldsymbol{v} \quad \text{for } j = 0, 1, \ldots, r-1,$$

and $(A - \alpha I)^r\boldsymbol{v} = \mathbf{0}$. Thus, the matrix representation of A with respect to the Jordan basis $\mathcal{B}$, called the **Jordan block**, is

$$J_r(\alpha) = \begin{pmatrix} \alpha & 1 & 0 & \cdots & 0 \\ 0 & \alpha & 1 & \cdots & 0 \\ 0 & 0 & \alpha & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & \cdots & \alpha \end{pmatrix}_{r \times r}.$$

More generally, if

$$V = \bigoplus_{i=1}^{k} V_i$$

is a direct sum of invariant and cyclic subspaces V_i with respect to A, and $\mathcal{B}_i$ is a Jordan basis for V_i with respect to A, then the ordered set $\mathcal{B} := \displaystyle\bigcup_{i=1}^{k} \mathcal{B}_i$ is a basis of V called a **Jordan basis** for V with respect to A. In this case, the matrix representation of A with respect to the Jordan basis $\mathcal{B}$, called the **Jordan normal form**, is

$$J = \bigoplus_{i=1}^{k} J_{r_i}(\alpha_i) = \begin{pmatrix} J_{r_1}(\alpha_1) & 0 & 0 & \cdots & 0 \\ 0 & J_{r_2}(\alpha_2) & 0 & \cdots & 0 \\ 0 & 0 & J_{r_3}(\alpha_3) & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \cdots & J_{r_k}(\alpha_k) \end{pmatrix}$$

where $\alpha_i \in \mathbb{C}$ and $r_i \geq 1$ for $i = 1, 2, \ldots, k$.

Theorem 8.4.13 Let V be a finite-dimensional vector space over $\mathbb{C}$ with $\dim V \geq 1$ and A be an operator on V. Then V can be decomposed as a direct sum of subspaces invariant and cyclic with respect to A.

Proof. Suppose there exists $\alpha \in \mathbb{C}$ and a positive integer r such that

$$(A - \alpha I)^r = 0, \quad (A - \alpha I)^{r-1} \neq 0.$$

Let $B := A - \alpha I$. We want to show that V can be decomposed as a direct sum of subspaces invariant and cyclic with respect to B. We will prove this by induction on $\dim V$.

Base case: if $\dim V = 1$, then B is a scalar operator, say $B = \lambda I$. Since $B^r = 0$, we have $\lambda^r = 0$, hence $\lambda = 0$ and $B = 0$. Thus, for any nonzero $\boldsymbol{v} \in V$, $V = \operatorname{span}\{\boldsymbol{v}\}$ is one-dimensional, invariant, and cyclic with respect to B.

Inductive step: now assume $\dim V > 1$ and that the result holds for all vector spaces of smaller dimension. If $\operatorname{im} B =: BV = \{\mathbf{0}\}$, then $B = 0$, so any basis $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ of V gives

$$V = \mathbb{C}\boldsymbol{v}_1 \oplus \cdots \oplus \mathbb{C}\boldsymbol{v}_n, \quad \text{where } \mathbb{C}\boldsymbol{v}_i = \operatorname{span}\{\boldsymbol{v}_i\} \text{ for } i = 1, 2, \ldots, n.$$

and each $\mathbb{C}\boldsymbol{v}_i$ is one-dimensional, invariant, and cyclic with respect to B. Hence we may assume $BV \neq \{\mathbf{0}\}$. Since B is nilpotent, $\ker B \neq \{\mathbf{0}\}$, and so $\dim BV < \dim V$ by the rank-nullity theorem. Also, BV is invariant under B. Hence, by the inductive hypothesis applied to $B|_{BV}$, we may write

$$BV = W_1 \oplus \cdots \oplus W_m,$$

where each W_i is cyclic with respect to B. Say $W_i = \operatorname{span}\{\boldsymbol{w}_i, B\boldsymbol{w}_i, \ldots, B^{r_i-1}\boldsymbol{w}_i\}$, with $B^{r_i}\boldsymbol{w}_i = \mathbf{0}$ and $B^{r_i-1}\boldsymbol{w}_i \neq \mathbf{0}$. Since $\boldsymbol{w}_i \in BV$, choose $\boldsymbol{v}_i \in V$ such that $B\boldsymbol{v}_i = \boldsymbol{w}_i$, and define $V_i := \operatorname{span}\{\boldsymbol{v}_i, B\boldsymbol{v}_i, \ldots, B^{r_i}\boldsymbol{v}_i\}$. Then each V_i is cyclic and invariant with respect to B.

Claim:

$$V' = V_1 + \cdots + V_m \text{ is a direct sum.}$$

Suppose $f_1(B)\boldsymbol{v}_1 + \cdots + f_m(B)\boldsymbol{v}_m = \mathbf{0}$, where $\deg f_i \leq r_i$. Applying B gives

$$f_1(B)\boldsymbol{w}_1 + \cdots + f_m(B)\boldsymbol{w}_m = \mathbf{0}.$$

Since $BV = W_1 \oplus \cdots \oplus W_m$, we get $f_i(B)\boldsymbol{w}_i = \mathbf{0}$ for every i. Writing $f_i(t) = c_{i,0} + c_{i,1}t + \cdots + c_{i,r_i}t^{r_i}$, the linear independence of $\boldsymbol{w}_i, B\boldsymbol{w}_i, \ldots, B^{r_i-1}\boldsymbol{w}_i$ implies $c_{i,0} = \cdots = c_{i,r_i-1} = 0$. Hence $f_i(t) = c_{i,r_i}t^{r_i}$. Substituting this back gives $c_{1,r_1}B^{r_1}\boldsymbol{v}_1 + \cdots + c_{m,r_m}B^{r_m}\boldsymbol{v}_m = \mathbf{0}$. But $B^{r_i}\boldsymbol{v}_i = B^{r_i-1}\boldsymbol{w}_i \in W_i$ and $B^{r_i-1}\boldsymbol{w}_i \neq \mathbf{0}$. Since $W_1 \oplus \cdots \oplus W_m$ is direct, each $c_{i,r_i} = 0$. Thus each $f_i = 0$, and therefore

$$V' = V_1 \oplus \cdots \oplus V_m.$$

Moreover, $BV_i = W_i$ for each i, so $BV' = BV$. Hence for any $\boldsymbol{v} \in V$, there exists $\boldsymbol{v}' \in V'$ such that $B\boldsymbol{v} = B\boldsymbol{v}'$, which means $\boldsymbol{v} - \boldsymbol{v}' \in \ker B$. Therefore $V = V' + \ker B$.

Now choose a subspace $U \subseteq \ker B$ such that $\ker B = (V' \cap \ker B) \oplus U$. Let $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_\ell\}$ be a basis of U. Then $V = V' \oplus U$. Indeed, since $V = V' + \ker B$ and $\ker B = (V' \cap \ker B) \oplus U$, we get $V = V' + U$; moreover, if $\boldsymbol{x} \in V' \cap U$, then $\boldsymbol{x} \in V' \cap \ker B$ and also $\boldsymbol{x} \in U$, so $\boldsymbol{x} = \mathbf{0}$. Hence

$$V = V' \oplus \mathbb{C}\boldsymbol{u}_1 \oplus \cdots \oplus \mathbb{C}\boldsymbol{u}_\ell.$$

Each $\mathbb{C}\boldsymbol{u}_j$ is one-dimensional, invariant, and cyclic with respect to B, since $B\boldsymbol{u}_j = \mathbf{0}$. Combining this with the decomposition $V' = V_1 \oplus \cdots \oplus V_m$ gives

$$V = V_1 \oplus \cdots \oplus V_m \oplus \mathbb{C}\boldsymbol{u}_1 \oplus \cdots \oplus \mathbb{C}\boldsymbol{u}_\ell,$$

which is the desired direct sum decomposition into invariant cyclic subspaces.
Finally, since $A = B + \alpha I$, every subspace invariant and cyclic with respect to B is also invariant and cyclic with respect to A and the scalar α. This completes the proof. □

Corollary 8.4.14 Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $A : V \to V$ be a linear operator. Then A can be written as $A = D + N$, where D is diagonalizable and N is nilpotent.

Proof. By the Jordan normal form theorem, there exists a basis of V with respect to which the matrix of A is a direct sum of Jordan blocks:

$$[A] = J_{k_1}(\alpha_1) \oplus \cdots \oplus J_{k_m}(\alpha_m).$$

Each Jordan block can be written as $J_{k_i}(\alpha_i) = \alpha_i I_{k_i} + M_i$, where M_i is the nilpotent Jordan block with zeros on the diagonal and ones on the superdiagonal. Define linear operators D and N by declaring their matrices in this Jordan basis to be

$$[D] = \alpha_1 I_{k_1} \oplus \cdots \oplus \alpha_m I_{k_m}, \qquad [N] = M_1 \oplus \cdots \oplus M_m.$$

Then $[A] = [D] + [N]$, hence $A = D + N$. Moreover, D is diagonal in this basis, so D is diagonalizable. Also, each M_i is nilpotent, and therefore the block diagonal matrix $[N] = M_1 \oplus \cdots \oplus M_m$ is nilpotent. Hence N is nilpotent. This proves the result. □

Theorem 8.4.15 Let $A, B \in \mathbf{M}_{n\times n}(\mathbb{C})$. Then A and B are similar if and only if they have the same Jordan normal form up to permutation of the Jordan blocks.

Proposition 8.4.16 Let $A \in \mathbf{M}_{n\times n}(\mathbb{C})$ and let

$$\chi_A(t) = (t - \lambda_1)^{m_1}(t - \lambda_2)^{m_2} \cdots (t - \lambda_k)^{m_k}$$

be its characteristic polynomial where $\lambda_i \in \mathbb{C}$ are the distinct eigenvalues for A with respective algebraic multiplicities $m_i \geq 1$ for $i = 1, 2, \ldots, k$. Let

$$\mu_i = \dim\ker(A - \lambda_i I) \quad \text{for } i = 1, 2, \ldots, k,$$

be the geometric multiplicity of λ_i for $i = 1, 2, \ldots, k$. Then there exist positive integers $\ell_{i,1}, \ell_{i,2}, \ldots, \ell_{i,\mu_i}$ such that

$$\ell_{i,1} \geq \ell_{i,2} \geq \cdots \geq \ell_{i,\mu_i} \quad \text{and} \quad \sum_{j=1}^{\mu_i} \ell_{i,j} = m_i,$$

and the jordan normal form of A is

$$\bigoplus_{i=1}^{k} \bigoplus_{j=1}^{\mu_i} J_{\ell_{i,j}}(\lambda_i).$$

Remark 8.4.17 Note, for each eigenvalue λ_i, the Jordan blocks corresponding to λ_i are determined by the integer partition (of m_i into μ_i parts)

$$m_i = \ell_{i,1} + \ell_{i,2} + \cdots + \ell_{i,\mu_i}.$$

Therefore, computing the Jordan normal form of A is equivalent to computing these integer partitions for each eigenvalue λ_i.

Theorem 8.4.18 Fix an eigenvalue λ_i of $A \in \mathbf{M}_{n\times n}(\mathbb{C})$ with algebraic multiplicity m_i and let

$$N_{i,j} := \dim\ker\left((A-\lambda_i I)^j\right), \quad j = 1, 2, \ldots, m_i,$$

and put $N_{i,0} := 0$. If the Jordan blocks corresponding to λ_i have sizes $\ell_{i,1} \geq \ell_{i,2} \geq \cdots \geq \ell_{i,\mu_i}$, then for each $j \geq 1$, we have

$$N_{i,j} - N_{i,j-1} = \#\{r : \ell_{i,r} \geq j\},$$

i.e., the difference $N_{i,j} - N_{i,j-1}$ is the number of Jordan blocks corresponding to λ_i of size at least j. Thus

$$\#\{r : \ell_{i,r} = j\} = (N_{i,j} - N_{i,j-1}) - (N_{i,j+1} - N_{i,j}) = 2N_{i,j} - N_{i,j-1} - N_{i,j+1}.$$

Remark 8.4.19 This gives us a way to compute the Jordan normal form of a matrix.

Method 8.4.20 Given a matrix $A \in \mathbf{M}_{n\times n}(\mathbb{C})$, we can compute its Jordan normal form by the following steps:

(1) Solve the characteristic polynomial

$$\chi_A(t) = \det(tI - A) = 0$$

to find the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$ of A and their respective algebraic multiplicities $m_1, m_2, \ldots, m_k$.

(2) For each eigenvalue λ_i, compute the nullities

$$N_{i,j} := \dim\ker\left((A-\lambda_i I)^j\right), \quad j = 1, 2, \ldots, m_i,$$

and put $N_{i,0} := 0$.

(3) For each eigenvalue λ_i, compute the number of Jordan blocks of size exactly j corresponding to λ_i by

$$\#\{r : \ell_{i,r} = j\} = 2N_{i,j} - N_{i,j-1} - N_{i,j+1}, \quad j = 1, 2, \ldots, m_i.$$

(4) For each eigenvalue λ_i, write down the block size

$$\ell_{i,1} \geq \ell_{i,2} \geq \cdots \geq \ell_{i,\mu_i}$$

and thus repeating the above process gives the Jordan normal form of A

$$\bigoplus_{i=1}^{k}\bigoplus_{j=1}^{\mu_i} J_{\ell_{i,j}}(\lambda_i).$$

Corollary 8.4.21 Suppose $A \in \mathbf{M}_{n\times n}(\mathbb{C})$ and its Jordan normal form is

$$\bigoplus_{i=1}^{k}\bigoplus_{j=1}^{\mu_i} J_{\ell_{i,j}}(\lambda_i),$$

where for each eigenvalue λ_i, the block sizes are $\ell_{i,1} \geq \ell_{i,2} \geq \cdots \geq \ell_{i,\mu_i}$. Then the minimal polynomial of A is given by

$$m_A(t) = \prod_{i=1}^{k}(t-\lambda_i)^{\ell_{i,1}}.$$

Remark 8.4.22 Note that the minimal polynomial of A is determined by the largest Jordan block corresponding to each eigenvalue λ_i. Thus, even if two matrices $A, B \in \mathbf{M}_{n\times n}(\mathbb{C})$ satisfy $m_A(t) = m_B(t)$ and $\chi_A(t) = \chi_B(t)$, it does not follows that A and B have the same Jordan normal form, and thus A and B may not be similar.